ELDERLY PEOPLE IN COOKING

MEET THE TEAM

Sinai Rhodes

Chief Operation Officer

As our COO, Sinai organised the workload of the group giving everyone their jobs and making sure they were done accordingly. He further helped the team with his work focused on competitor products found throughout the market and opportunities that arose from issues within those products.

Emmanuel Irechukwu

Chief Creative Officer

Emmanuel led the group in regards to the layout and styling of the portfolio creating a colour scheme based off existing brands we'd like to work with and adding illustrations and visual cues such as graphs and charts. He also helped to create user persona's and got the key findings and insights from them.

Alexander Wilton

Chief Technical Officer

Our CTO, Alex, led the group in the disassembling and teardown of the existing product, both separating it into its main components and then weighting, analysing and measuring them to work out what material they were made of and how they would have been produced to give us a clear comparison and multiple insights.

Francesco Rastelli

Chief Information Officer

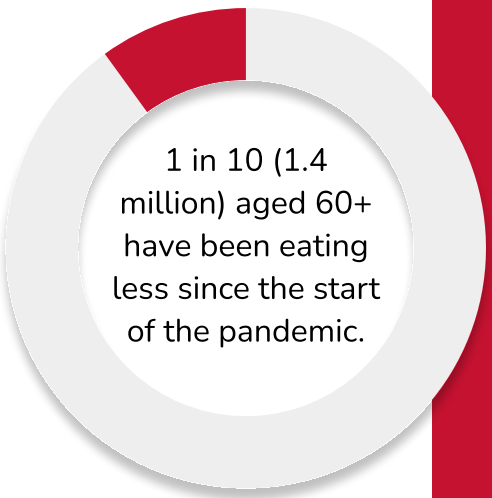
As the Chief Information Officer, Francesco led the survey and interview stage ensuring that a wide range of people were talked to and collecting the data so that it could be passed on to the rest of the team allowing them to develop insights which influenced the final products.

INTRODUCTION

Background

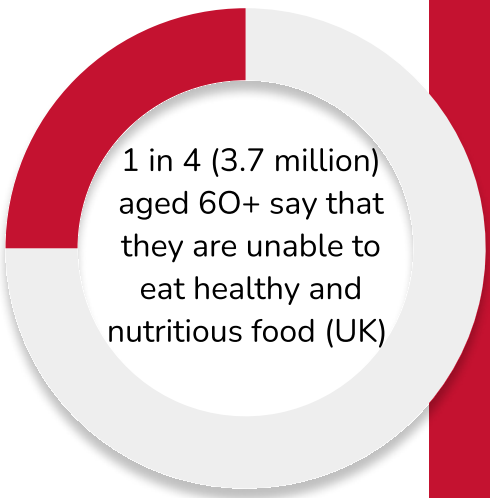
General Information

Cooking, a universal activity that transcends cultural and demographic boundaries, and is a crucial aspect of daily life. While people of all ages and backgrounds engage in cooking, certain age groups, particularly those over 60, face unique challenges. These can greatly vary allowing us to have a broad range of possible problems to tackle and solve for the betterment of the elderly cooking experience.



Facts & Figures

The two figures on the right have disturbingly given us an idea of quite how bad the problem can be for the elderly. It is estimated that approximately one in ten individuals in the elderly demographic are malnourished or at risk of malnutrition in the UK. This highlights the critical importance of addressing cooking-related issues among the elderly population.



Approach And Rationale

Initial Research
We began by researching cooking around the elderly and getting an idea of potential issues they could have. This was further developed by early interviews which allowed us to gain a better understanding of the topic.

User Research
Interviews were conducted which allowed us to identify the main issues to do with cooking for elderly and to build a map of their struggles. This led to many insights which were then used to find potential product opportunities.

Market Analysis
Looking at existing products within the current market tailored to both the elderly and to the average user allowed us to gain insights into both potential product ideas and ways to improve existing products.

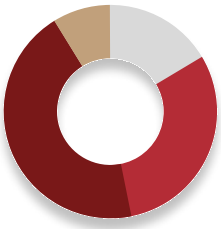
Conception
Solutions could thus be developed based off of the research and market analysis that were conducted. The insights we got from the previous steps thus highly impacted what our final product options would be.

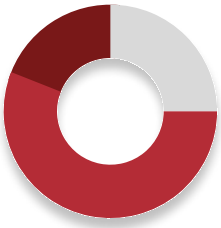
USER RESEARCH

Identifying Our Users

To identify our user group, we needed to work out what issues elderly people are facing in the home when undertaking cooking. While most of the group had access to grandparents which could be interviewed, it was still good to be able to gain access from external sources. For that, we created a survey which we spread around and used to both gain quantitive and qualitative data and to see if we could find any extra people willing to be interviewed.

Survey Results





Important survey quotes:

1 - "He eventually caved in with mixers and such when he got tired easily."

13 - "I think as she gets older she will not have the energy or the motor skills to be cooking as much as she is at the moment."

32 - "As grandparents age, they are less and less able to do more complex recipes, or forget their recipes coz usually they do them from memory"

33 - "Their cooking habits will decrease with age as they will have less energy to cook"

What challenges do the elderly face while cooking?

The elderly face lots of different challenges while cooking due to illness and other factors afflicting them. Many responses talked about how even though they can still cook, the time taken for the same recipes has increased greatly due to "things like carrying pots and stirring [becoming] harder". Forgetfulness further plagues the elderly, one respondent said that "she used to forgetting burning and losing track of time" while another has issues with remembering and "gathering all the ingredients" which leads to "burning food or overcooking".

Ideal devices to make cooking easier

For many respondents, they want to have a "more efficient dicing instrument" to help them with cutting things, however, it is also important for it to be easy to clean. The other issue they seem to have faced is with the height of objects "to reduce strain on the neck", so being to adjust the level of stoves and ovens is important.

RESEARCH PLAN

The Aim:

Within this research, we aimed to find the issues that the elderly have to deal with in cooking by analysing surveys and user interviews that we conducted. This is to lead to the creation of a product that would improve the experience of cooking for our user group.

Interview Methodology:

The interviews were conducted to try and gain a deeper understanding of the issues that plague our users. While the survey gave us answers to a basic set of questions, the interviews showed us the emotions that our user group feels when cooking and helped us to develop multiple insights from them.

Desk Research

Desk research focused on finding the demographics and key information around the topic. Gives an oversight of the problem and potential issues we may have.

Survey

Survey gave us preliminary information and access to participants for the interviews. Allows us to have users answer a wide range of basic questions in order to find recurring challenges that they face.

Interviews

Interviews were run with a range of users with focus being placed on the ones with highly differing issues.

- **User Personas**
- **Key Findings and Insights**

Comparison Of Competitor Products

Looking at existing products helped us find design opportunities through issues that we found in the existing products.

- **Teardown Of Competitor Product**

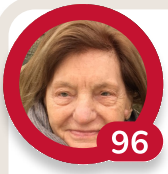
USER INTERVIEWS

Birgitta Wiklund

Brigitta is a typical Swedish grandma who has unfortunately been windowed. This has led to her not enjoying cooking much anymore, as she found the joy in cooking in doing it for others:

"In the summers when daughter and grandchildren come it becomes more fun because you cook for more people."

Unlike most elderly people, Brigitta has embraced new technology including things like air fryers which she says helps her a lot with cooking things much quicker and more efficiently.



Piera Bragadin

Pierina is unfortunately afflicted by Alzheimer's, and it has developed up to the point of where she is now unable to do even the most basic tasks. As the illness developed, she started to forget things on the stove or in the oven, and later on would often be found with a chair next to the stove or sink using it whilst she cooked. It also led to losing a myriad of delicious family recipes. Sever pains in her fingers further made the cooking process difficult as she became unable to grip things and would try to massage other people's hands thinking they were hers.

Anthony Walsh

Tony reminisces about his routine of cooking daily. After having returned home from battling cancer in hospital, after a year he is still yet to make his return to the kitchen.

A Jamaican born and bred he took such pride in preparing West Indian meals for his family. Traditional methods appeal to him, as he mostly uses a knife, with technology having not significantly influenced his cooking habits, preferring authenticity over convenience. He enjoys cooking with his daughter when he gets the chance and does grocery shopping weekly but often would forget an item or two .

Patricia Pridmore

Pat is a retired academic who cooks for her husband, John Pridmore due to him being older and less able. She has recently undergone knee surgery on both legs which created problems with any moderate-heavy lifting, especially moving weight from low to high places. She intimated that feelings of frustration arose in her during meal preparation since she must cook healthy double portions thrice a day even though she herself is elderly. The frustration mainly stems from fatigue brought on by cooking - particularly dinner, which is relatively late in the day.

Key Insights

A lack of strength leads to an inability to cook effectively and getting tired quickly thus removing enjoyment in the process.

Forgetfulness leads to multiple issues when forgetting things on the stove, when shopping and also with forgetting and ultimately losing recipes.

Loneliness in a large number of elderly leads to them not enjoying the cooking process anymore, with many only liking to cook if others are around.

USER PROFILES

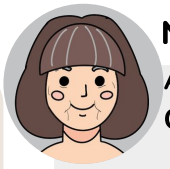
User Persona's

Family-Focused Fiona

Age:
Gender: Female

Cooking Style:

Only finds joy in cooking for family gatherings where she can showcase elaborate meals and enjoys the communal aspects of cooking



Nostalgic Nancy

Age:
Gender: Female

Cooking Style:

Once enjoyed creating intricate and memory-rich meals, now adapts to simpler cooking due to changes in physical stamina and cognitive abilities.

Energy-Efficient Ed

Age:
Gender: Male

Cooking Style:

Acknowledges a decrease in energy and mobility, which influences their approach to cooking, leading to quicker and less demanding meal prep

Needs

Tools and appliances that help in creating elaborate meals for family gatherings; recipe resources to expand her culinary repertoire for these occasions.

Kitchen appliances and tools that simplify the cooking process. Clear accessible recipes to aid with memory challenges.

Minimalistic cooking processes with straightforward steps, and safety features designed to accommodate for their changing capabilities.

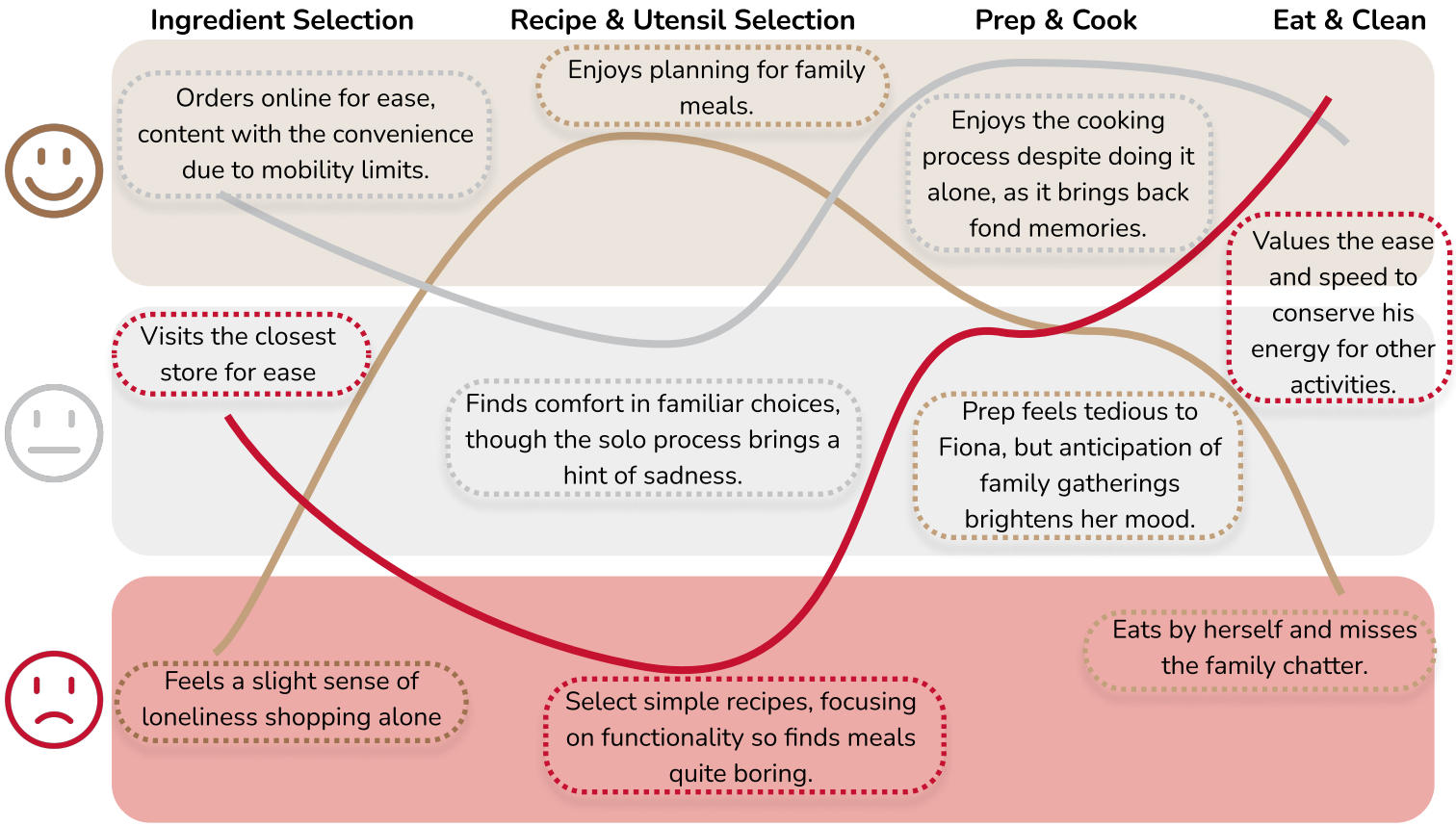
Goals

To maintain the joy of cooking for family, creating memorable and delicious meals that bring everyone together.

To recreate the cherished flavours of the past with minimal effort and continue to enjoy cooking as a comforting and familiar activity

To maintain independence in the kitchen, ensuring that meals can be prepared safely and effortlessly.

User Journey Maps



KEY FINDINGS & INSIGHTS

07

01 Elderly individuals exhibit a wide range of physical and cognitive capabilities, affecting their ability to cook and enjoy the process.

A shift in cooking ability is noted as elderly individuals age, with some maintaining strength and others facing difficulties such as **lifting** or **stirring** due to **diminished grip strength**, as is common in the **70-79 age demographic.**

02 There's a growing need for ergonomic kitchen tools and smart technology that cater to the changing physical and cognitive abilities of elderly users.

The call for ergonomic and technological aids is clear, with a preference for tools that alleviate physical strain, like **"efficient dicing instruments"**, and stoves that adjust in height to accommodate limitations in mobility and energy levels in the **80-89 and 90+** age brackets.

03 Loneliness and emotional well-being are closely tied to cooking activities, impacting the elderly's engagement and fulfillment.

Elderly cooking experiences are deeply intertwined with their social lives. For instance, **individuals like Brigitta**, who was recently widowed, find the joy of cooking **significantly diminished** when done in solitude, indicating a profound sense of loneliness.

Opportunities

Diverse Capability Spectrum

HMW: develop cooking tools that cater to the diverse physical abilities of elderly cooks, ensuring everyone can prepare meals safely and with ease?

Tech & Ergonomic Solutions

HMW: incorporate ergonomic design and technology into cooking aids to better suit the changing needs of elderly users?

Social Dynamics & Cooking

HMW: enhance the cooking experience to address the social needs of elderly individuals, making it more engaging and less isolating?

PRODUCT OPPORTUNITY

Summary Of Opportunity

Combining the opportunities found into one holistic approach addresses the complex needs of elderly individuals in the kitchen. The aim is to create solutions that are physically accessible, cognitively supportive, and socially enriching. By doing so, we acknowledge that effective cooking aids must enhance ease of use, foster social engagement, and adapt to ergonomic requirements, thereby rekindling the joy and sense of purpose that cooking can bring to elderly users' lives.

Diverse Capability Spectrum

Social Dynamics & Cooking

Tech & Ergonomic Solutions

How Might We:

"design cooking aids that not only reduce stresses for elderly users during common cooking tasks but also infuse purpose into their cooking process, thereby enabling elderly individuals to enjoy cooking once again?"

Design Specification

1. Products must be lightweight with easy-to-grip handles and controls that are simple to operate, accommodating a range of mobility and strength levels.
2. The design should feature straightforward, clear interfaces with large buttons or touchscreens and audible cues to assist users with varying cognitive abilities.
3. Aids should offer capabilities for connectivity, enabling users to share experiences or receive support, fostering a sense of community.

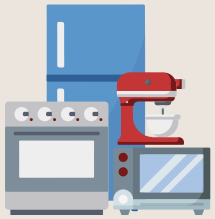
MARKET AND COMMERICAL

Initial Research

As the elderly face declining health, their participation in cooking decreases, affecting their nutrition and well-being. Cooking plays a vital role in promoting independence and this shift in household dynamics can weigh heavy on both nutritional and mental well-being. The product categories for this demographic can be split into the following 3 groups:

Preparation Tools

Cooking Appliances



Motivation and Reminder Tools

Market Research

In 2024, the revenue in the **Small Kitchen Appliances** market worldwide amounts to

$133.6B

This is projected to grow from 2024 to 2028 with a **CAGR** of

%5.03

Frequency of people 60+ who cook

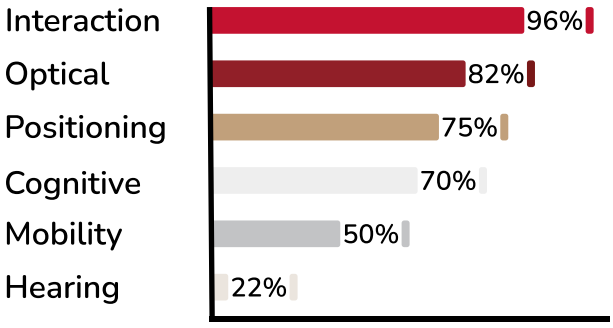
The global **Elderly and Disable Assisted Devices** market in 2022 was valued at

$27B

This is set to grow from 2023 to 2032 with a **CAGR** of

%5.70

Percentage of people 60+ using different types of Assistive Devices



Key Trends Over Time

Online grocery delivery and **meal kit** providers became more popular.

Smart technologies and **AI integration** into existing product to add personalisation and adaptability

COVID-19 caused the **elderly to drop out of the kitchen** due to illness.

Touch-sensitive and **voice-activated devices** were introduced to help reduce contact

Compact and **multi-functional products** to maximise storage and provide versatility.

COMPETITOR PRODUCTS

Electric Jar/Can Opener

£20 - £60

Pros
- Compact for storage
- Easily Accessible (One-touch)
- Safe for elderly
- Efficient

Cons
- Temperamental (doesn't open all)
- Noisy
- Single use (only opens jars/cans)

Voice-Controlled Microwave

£80 - £140

Pros
- Personalised.
- Hands-free cooking
- Accessible for visually impaired
- Multi-functional

Cons
- Difficult to set up
- Learning curve for elderly users
- Very expensive for a microwave

Automatic Pot Stirrer

£10 - £40

Pros
- Hands-free cooking
- Compact for storage
- Easily Accessible (One-touch)

Cons
- Damages Pot
- Gets stuck in thicker liquids
- Not very heat resistant

Vibrating Alarm Watch

£80 - £140

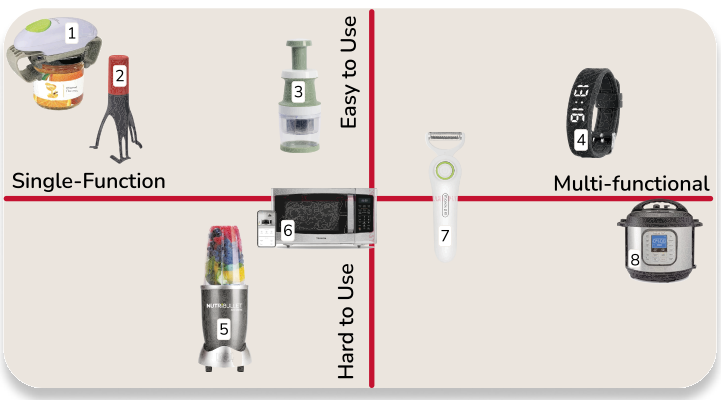
Pros
- Multifunctional (alarms for food, water, pills etc.)
- Long Battery Life
- Rechargeable

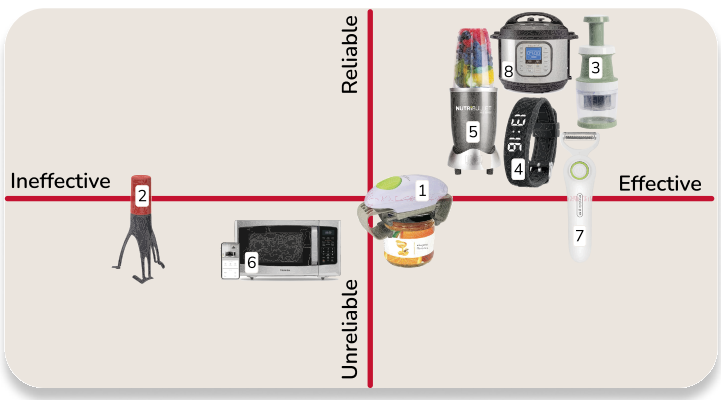
Cons
- Uncomfortable
- Limited visual displays
- Not waterproof

	Ease of Use	Versatility	Confidence-boosting	Effectiveness	
Electric Jar/Can Opener	5	2	3	4	14/20
Voice-Controlled Microwave	3	4	4	3	14/20
Automatic Pot Stirrer	5	1	2	2	10/20
Vibrating Alarm Watch	4	5	4	4	17/20

The matrices below categorise 8 products into the four main characteristics of an ideal product, ease of use, number of functions, reliability and effectiveness.

1 Can Opener
2 Pot Stirrer
3 Chopper
4 Alarm Watch
5 Blender
6 Talking Microwave
7 Peeler
8 Pressure Cooker





DESIGN OPPORTUNITIES

From analysis of competitors manufacturing kitchen appliances and reminder devices for the elderly, the following market gaps and design opportunities have been identified:

User Interaction and Accessibility

Elderly users struggle enough when changing habits and particularly in the kitchen when they have performed tasks one way for decades. As a result, it is imperative products are as simple and accessible as possible.

Multi sensory Alerts

- Implementation of these can ensure users, regardless of sensory impairment, receive alerts, as seen by the vibrating alarm watch.
- This allows us to cater to as many disabilities and impairments as possible.

Diverse Input Options

- Large, tactile buttons reduce the complexity of operating devices as the elderly can struggle with small, touch-sensitive controls.
- Voice-operated products are effective in operation but difficult to set up.

Adaptability and Customisation

Personalisation of tools and functionalities allows users to make their product feel unique to them. The products ability to adapt to the users habits allows their user experience to get better over time as it becomes more tailored to them.

Ergonomic Designs

- Tailoring products to accommodate different grip strengths ensures users with physical limitations, can use devices comfortably.
- This is vital for enhancing usability as seen with the jar/can opener.

Multi-functionality

- Offering modular attachments, a single device can serve multiple functions, expanding its utility.
- This is seen with the electric peeler or the alarm watch providing many alarms for different occasions.

Enhanced Safety and Functional Aesthetics

As people grow in age motor skills can decline leading to more frequent accidents. Safety is a top priority in addition to ensuring the product is aesthetically enticing and discreet to not only look good but in turn boost confidence.

Safety Features

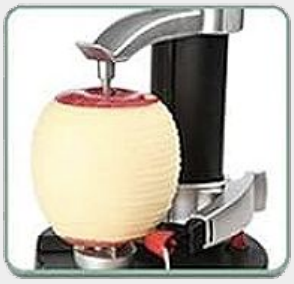
- Integrating auto-off features can significantly reduce the risk of accidents caused by overheated appliances or forgotten turn-offs.
- This can be important to help combat forgetfulness and cognitive impairments.

Discreet Design

- Discretion without compromising functionality. Users should be able use assistive products without feeling out of place.
- Unobtrusive design can ensure the user feels comfortable and confident.

PRODUCT OVERVIEW

Automatic Peeling Machine



Functions

The electric potato peeler by Aibrand offers a convenient alternative to the oftentimes arduous process of hand peeling assorted vegetables and fruits. The user simply inserts the lower end of the foods longest axis onto a set of anchor spikes and then insert the upper pin into its top. the blade arm is then unlocked and relocated to just above the upper pin. Once activated, the arm now carves a diagonal line downwards, removing about a millimetre from the entire surface except for a small region about the poles.

First Impressions

Appearance

- ✓ Modern, polished appearance.
- ✓ Sturdy and stable bass.
- ✗ Very light and cheap feeling due to almost entirely ABS build.

Result

- ✓ One handed operation.
- ✓ Removes most of the skin.
- ✗ Slower than by hand.
- ✗ Excessive amount of flesh removed.
- ✗ Skin remains around anchors.

Market

- ✓ Reasonable demand with thousands of purchases on Amazon alone
- ✗ Marketed to those who would not realistically need or want it, leading to polarised reviews.

TEARDOWN

The Board

Notable Features

ERGONOMIC	INTERNAL	FUNCTIONAL
• Textured paddles give good grip • Blunt spikes are not dangerous • Affermative feedback from buttons etc • Fairly obvious operation, easy and fast to learn	• Minimalistic, functional electronics • Metal axles have good longevity • All electrical components housed in lower body • Single motor actuates multiple things using a gear train.	• Battery or mains operable (integrated battery box) • Consistent performance • Easily activable kill switch for safety • Capable of removing a variety of skins • Included magazine of spare blades for fast maintenance

PARTS LIST, DETAILS

Exhaustive List:

Body				
PART	NUMBER	MASS (g)	MATERIAL	MANUFACTURE
Upper Base	1	63	ABS	Injection Moulding
Lower Base	1	85	ABS	Injection Moulding
Stem	1	69	ABS	Injection Moulding
Silicon Seal	1	13	Silicone	Molding
Battery Box	1	13	ABS	Injection Moulding
Battery Lid	1	5	ABS	Injection Moulding
Charger Pin Face	1	12	ABS	Injection Moulding
Charger Body	1	18	ABS	Injection Moulding
Gearbox Face	1	5	ABS	Injection Moulding
Pillar Top Buffer	1	7	ABS	Injection Moulding
Buffer Cap	1	2	ABS	Injection Moulding
Blade Storage Lid	1	4	ABS	Injection Moulding
Blade Padding	1	0.5	Polyethelyne	Injection moulding
Technical				
Pillar Screw	1	28	ABS	Injection Moulding
Slider	1	16	ABS	Injection Moulding
Top Pivot	1	22	ABS	Injection Moulding
Slide Adjuster	1	10	ABS	Injection Moulding
Lower Pivot	1	8	ABS	Injection Moulding
Black Drive Gear	2	5	ABS	Injection Moulding
White Drive Gear	1	5	ABS	Injection Moulding
Large Ratio Gear	1	2	ABS	Injection Moulding
Medium Ratio Gear	1	1	ABS	Injection Moulding
Small Ratio Gear	1	0.5	ABS	Injection Moulding
Primary Drive Gear	1	0.2	ABS	Injection Moulding
Motor Complex	1	35	Stainless Steel	Casting
Blades	3	1.5	Stainless Steel	Stamping
Charger Cable	1	19	Copper	Drawing
Blade Housing	1	4	ABS	Injection Moulding
Pivot Adjuster	1	3	ABS	Injection Moulding
Internal Wiring	1	5	Copper	Drawing
Plug Power Board	1	9	Silicone	Molding
Battery Terminal	4	0.9	Copper	Drawing
Big Spring	1	0.6	Stainless Steel	Casting
Small Spring	1	0.3	Stainless Steel	Casting
Big Gear Axle	2	0.3	Stainless Steel	Casting
Small Gear Axle	1	0.2	Stainless Steel	Casting
Battery Ribbon	1	0.1	Polyester	Weaving
Assembly				
Slide Retainer Pin	1	2	Stainless Steel	Casting
Slide Pivot Pin	1	3	Stainless Steel	Casting
Wide Head Screw	5	0.6	Stainless Steel	Casting
Long Screw	4	0.7	Stainless Steel	Casting
Nuts	2	1	Stainless Steel	Casting
Regular Screw	7	0.4	Stainless Steel	Casting
Short Wide Screw	2	0.3	Stainless Steel	Casting
Short Normal Screw	2	0.2	Stainless Steel	Casting
Small Washer	1	0.1	Stainless Steel	Casting
Large Washer	1	0.1	Stainless Steel	Casting

Overview:

As is typical of many cheap modern electrical appliances, ABS is liberally employed in almost every aspect of the design. This grants the automatic peeler the benefit of being light, sturdy and also easily cleanable. Stainless steel components are sporadically dotted across the assembly as well with most of the critical and kinetic components needing to withstand localised stresses.
Electrical conduits house copper wiring for power transmission.
A large silicone seal protects the working parts from environmental threats like moisture.

Comfort:

ABS, being a smooth insulator, is pleasant to handle. It is also non-porous and therefore has favourable hygienic properties and is easy to clean after use.

Longevity:

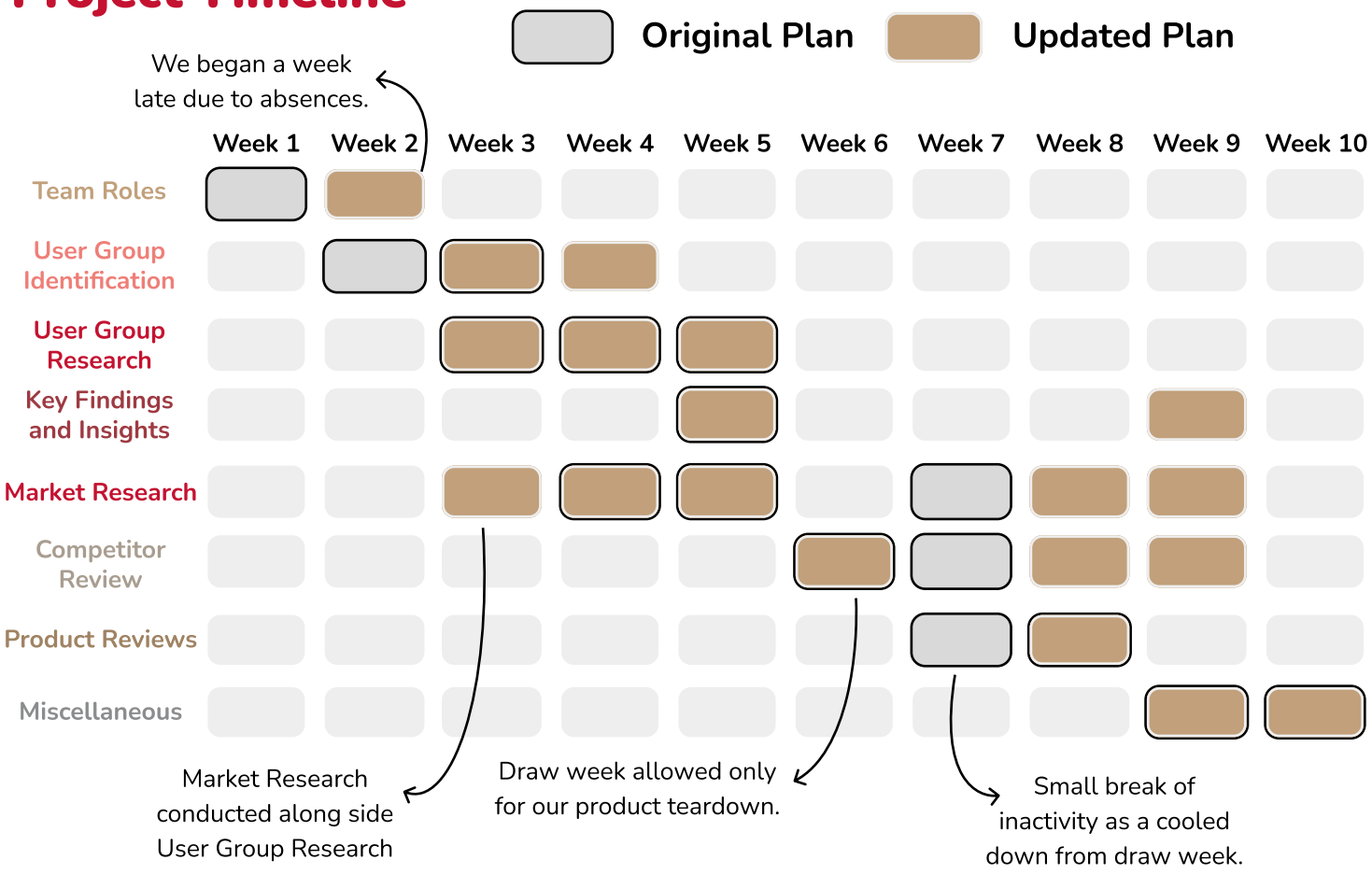
This product promises to last a long time due to its ABS and stainless constituents. The greatest weakness arguably is the plastic screw threads which could fail under extraordinary torsional or tension forces.

Assemblability:

The purposeful eschewal of adhesives permits the peeler to be comprehensively stripped down to its most fundamental parts. Only the externally sourced plug and electronics are not easily dismantlable.

PLAN - EVOLUTION

Project Timeline



Timeline Review

Each team member had one role, COO, CCO, CTO and CIO. These were assigned democratically.

We struggled to settle on a user group and so conducted user and market research on 2 separate markets before deciding on Elderly People in Cooking. We then ran an online survey to try an understand our general user's habits.

Our User Research followed up from our survey where we each contacted one respondent to conduct a 1-to-1 interview to further assess problems our users face. As a result of these, we formed user profiles to categorise our types of user before leading into our key insights and product opportunity. These insights helped us hone in and define our How Might We? for further development.

Market Analysis proved difficult gain an understanding of the market size of both Elderly Assistive Devices and Kitchen Appliances, analysing the trends and overlaps of both.

Analysis of competitor products allowed us to identify pros and cons of different products and designs. This helped identify market gaps and opportunities to implement into our own ideation.

The group partook in a product teardown of a provided product, the automatic peeler. This involved the full dissection of the device to analyse the material and production process of each component.

Miscellaneous pages including cover page, plan reflection and MCS were completed in final weeks.

PLAN - CONTRIBUTION

Emmanuel

Responsible for: user profiles (p6), key findings and insights (p7), production opportunities (p8), multiple choice survey (p17).

Alex

Responsible for: peeler overview (12), teardown (13), parts list + details (14) , contributions (16).

Francesco

Responsible for: meet the team (1), user background (3), research plan (4), user interviews (5).

Sinai

Responsible for: market research (9), competitor products (10), design opportunities (11), plan evolution (15).

Work was assigned and distributed via a team Whatsapp group by the COO. This entailed the determining individual workloads and time constraints. The CIO compiled a collective range of interviews and unearthed a range of useful insights from these with which the team was able to refine the spec that a potential solution would have to adhere to. Our CCO analysed existing, successful brands and formulated a document theme and colour scheme based on these as well working on general cohesion of pages. The CTO worked on the practical aspects of relevant product disassembly and analysis.

What next?	Using the information gained from the team correspondence, four individual concepts will be individually developed and appraised.

MULTIPLE CHOICE SURVEY

Concept 1 - Legacy

Concept 2 - Mashomatic

Concept 3 - StirMaster Pro

Concept 4

01 Which concept do you think will require the least physical effort to use?

- ◯ Concept 1
- ◯ Concept 2
- ◯ Concept 3
- ◯ Concept 4

02 Which concept do you find most intuitive and user-friendly?

- ◯ Concept 1
- ◯ Concept 2
- ◯ Concept 3
- ◯ Concept 4

03 Which concept would best assist with maintaining a clean and organized kitchen space?

- ◯ Concept 1
- ◯ Concept 2
- ◯ Concept 3
- ◯ Concept 4

04 Which concept do you feel would have the most positive impact on your daily routine?

- ◯ Concept 1
- ◯ Concept 2
- ◯ Concept 3
- ◯ Concept 4

05 Considering your mobility and dexterity, which concept seems most accommodating?

- ◯ Concept 1
- ◯ Concept 2
- ◯ Concept 3
- ◯ Concept 4

06 Which concept would you trust to enhance your independence at home?

- ◯ Concept 1
- ◯ Concept 2
- ◯ Concept 3
- ◯ Concept 4

07 Which concept appears to offer the best value for money?

- ◯ Concept 1
- ◯ Concept 2
- ◯ Concept 3
- ◯ Concept 4

08 Which concept do you believe is most likely to reduce the feeling of loneliness by keeping you engaged?

- ◯ Concept 1
- ◯ Concept 2
- ◯ Concept 3
- ◯ Concept 4

09 For those with arthritis or hand pain, which concept seems most suitable?

- ◯ Concept 1
- ◯ Concept 2
- ◯ Concept 3
- ◯ Concept 4

10 Which concept do you think will best support cognitive challenges, such as remembering to perform certain tasks?

- ◯ Concept 1
- ◯ Concept 2
- ◯ Concept 3
- ◯ Concept 4

USER INSIGHTS

The problem:

Looking at our research, we found many issues with elderly by conducting both surveys and interviews with them. Thus, by looking at the insights we gained, we ended up reaching the following HMW:

How might we design cooking aids that not only reduce stresses for elderly users during common cooking tasks but also infuse purpose into their cooking process, thereby enabling elderly individuals to enjoy cooking once again?

This was done to deal with both a severe and recurring issue we found where elderly people said they did not enjoy cooking anymore due to things like loneliness, to trying to tackle the issue that almost all of them face where they lack the strength to deal with the more basic tasks.

Survey Insights:

The survey gave us the initial ideas of what emotions the elderly undergo when cooking and issues they face.

1. Cooking habits decrease with age as their energy and motor skills deteriorate.
2. Remembering to gather all ingredients for a recipe can lead to them messing up.
3. Many would like a more efficient dicing instrument.
4. Strain on legs, back and neck affects many elderly people, especially with low countertops.

Interview Insights:

The interviews further deepened our understanding of the emotions and frustrations faced while helping us divide the issues they face into 4 key groups:

1. Those dealing with a worsening illness such as Alzheimer leading to forgetfulness.
2. Those recovering from an illness, and thus lack strength and motivation.
3. Those who lack the endurance and struggle to stand for long periods of time.
4. Those who struggle with loneliness and thus don't enjoy cooking due to it.

Key Insights:

A lack of strength leads to an inability to cook effectively and getting tired quickly thus removing enjoyment in the process.

Forgetfulness leads to multiple issues when forgetting things on the stove, when shopping and also with forgetting and ultimately losing recipes.

Loneliness in a large number of elderly leads to them not enjoying the cooking process anymore, with many only liking to cook if others are around.

Francesco
19

MARKET INSIGHTS

Issues with existing products:

- Many products are simply unaffordable to the average user, especially with the growing costs of living.
- Lack of multi-function-ability decreases utility and thus desirability.
- Lack of interactive cues leads to uncertainty as to wether the product is working or not.
- Poor material and build quality leads to low durability and so a poor product
- Little to no thought given to the safety of the user
- A high learning curve or overcomplexity leads to a low ease of use.
- Low comfortability due to bad ergonomics.

Market Opportunities

User Interaction and Accessibility

Elderly users struggle enough when changing habits and particularly in the kitchen when they have performed tasks one way for decades. As a result, it is imperative products are as simple and accessible as possible. This can be done by:

Multi-sensory Alerts

- Ensures users receive alert, especially compared to visual or auditory alarms.

Diverse Input Options

- Large tactile buttons and voice activated devices reduced complexity for users.

Adaptability and Customisation

Personalisation of tools and functionalities allows users to make their product feel special and unique to them. The products ability to adapt to the users habits allows their user experience to get better over time as it becomes more tailored to them.

Ergonomic Designs

- Helps users with weaker grip strengths hold onto products comfortably enhancing usability.

Multi-functionality

- Offering modular attachments allows a device to have multiple functions.

Enhanced Safety and Functional Aesthetics

As people grow in age motor skills can decline leading to more frequent accidents. Safety is a top priority in addition to ensuring the product is aesthetically enticing and discreet to not only look good but in turn boost confidence.

Safety Features

- Auto-turn off features and ergonomic designs reduce the risks of injuries.

Discreet Design

- Designs should be able to be used without making the users feel out of place.

IDEATION

Product Aims:

Looking at the HMW, insights that we got from the surveys and interviews, and the market research where we found issues with existing products and opportunities that arose form them, we found a list of ideal goals that our products should aim for:

The product should be easy to both move around and use requiring a minimal amount of strength and dexterity.

The product should aid users and support them through both degradation into illness and recovering from it.

The product should not have a high learning curve allowing non tech-savy users to use it.

Safety factors should be looked at within the product to reduce the risk of injury.

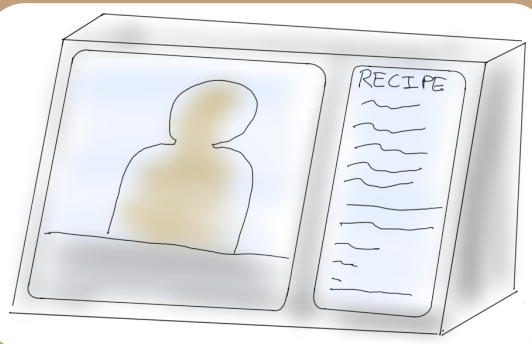
The product should bring some form of enjoyment back to cooking by either creating connection or reducing stress.

The designs should not make a user feel “less-abled” but instead discretely aid the user.

Potential Products

1. Legacy Scale

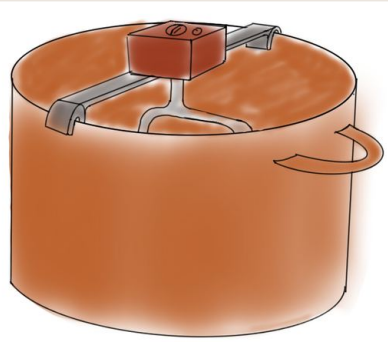
This product would be used to record traditional recipes down by weighing and recording each ingredient. Then, as illnesses develop or the recipe wants to be passed on, it can be used the opposite way by giving visual and auditory feedback to the user (using the LEDs) allowing them to recreate recipes even if they don’t remember them.



2. Cook Connect

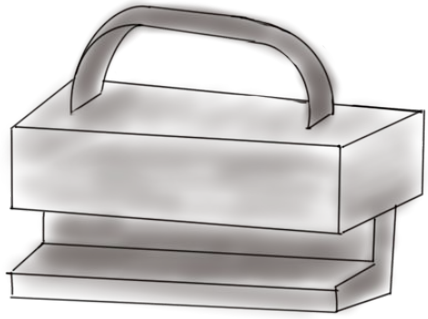
In order to both reduce loneliness and pass on traditional recipes, this product would allow two people to call each other and cook the same recipe at the same time. This could be used to both teach the younger generation and to add purpose to the elderly’s cooking by adding a form of connection and higher purpose to it.

IDEATION



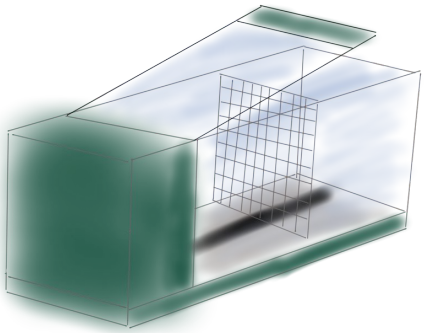
3. Tempo Mix

Tempo Mix would be a mixer that could be placed on top of any pot and would use a timer which would continuously mix until the time runs out, after which it would start an alarm. This would ensure that food is kept stirred reducing the chances of burning and removing physical stress from the user.



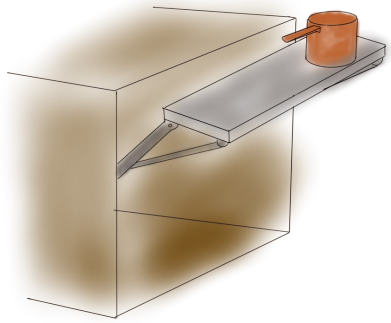
4. Electro Grip

The ElectroGrip has been envisioned to aid users in gripping objects by tightening on handles or holds and offering much easier to user and grip straps which do not require as much finger strength or dexterity to use. The product would simply be placed on a handle and activated by pulling on the straps to lift them.



5. Easy Dice

The aim of this product is to make the repetitive task of chopping things easier. It would use a set of thin metal wires on a rail system to cut through vegetables, fruits and other products. To ensure safety, it would only work when the lid is closed and could be turned on by putting weight on it making it easier to use.



6. Easy-lift Drawer

This product makes the task of raising pots and pans from low down places a lot easier by lifting it to countertop height for the user helping the elderly deal with things like back and leg pain through the avoidance of having to bend over and lift with the back. This could be moved using a simple pulley and motor system.

CONCEPT SELECTION

Benchmarking:

	Manuverability	Long Term Support	Learning Curve	Safety	Enjoyment and Connection	Discreetness	Total (/30)
1. Legacy Scale	02	05	01	05	04	04	21
2. Cook Connect	02	02	01	05	05	03	18
3. Tempo Mix	03	04	04	03	02	04	20
4. Electro Grip	05	02	04	04	01	02	17
5. Easy Dice	03	04	03	02	02	03	17
6. Easy-lift Drawer	01	03	04	02	01	02	13

The Benchmarking criteria were based off the product aims that can be found on Ideation Page 3. Each criteria was marked out of five and the total were then summed up a the end to give us a roughly quantative result on the performance of the products.

User Feedback On Products:

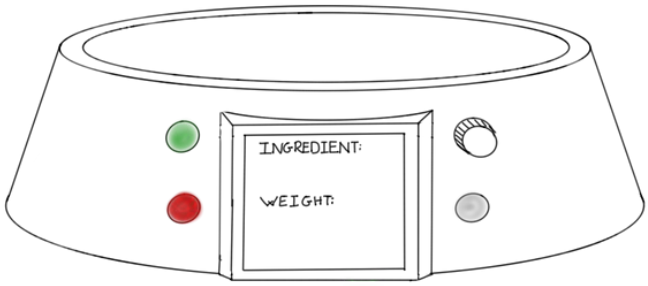
To fully understand what product to bring forwards in development, I had to go back to the users and talk to them about what they liked and did not like about each product.
The products who got the least praise were the Electro Grip (due to worries with it simply slipping off) and the Cook Connect (due to a lack of usefulness being cited). Both the Easy-lift Drawer and Legacy Scale were liked but neither were seen as very useful, although some credit was given to the ability of the scale in the long term support, especially by Pierina's group; who still reminisce of her old recipes. The last two were the most liked: The Easy Dice has a clear usefulness, with the only worrying being with it possibly breaking when cutting through hard vegetables like potatoes; meanwhile, the Tempo Mix was seen as extremely useful, especially by one who said that she'd "love it for making polenta".

What product to bring forwards then?

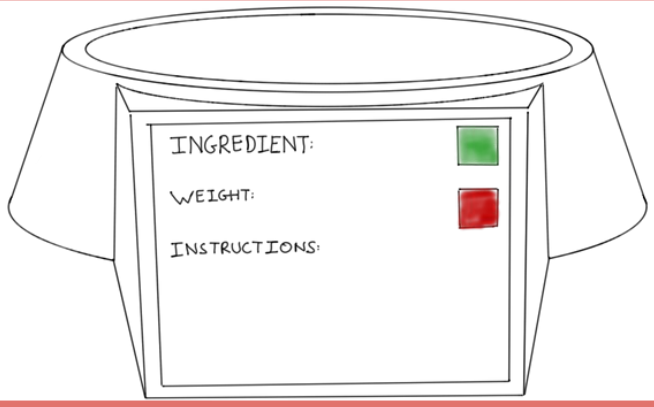
Looking at the overall scores from the benchmarking, the first product to go is the Easy-lift Drawer; with the Electro Grip following suit due to user feedback and also low score. Whilst, the Easy Dice was liked by the users, in the end it's benchmarking score was too low to get carried forward. The Cook Connect instead is being rejected due to the simple reason that it could be made as an app and be a better product that way. In the end, the last two products are the Legacy Scale, and the Tempo Mix, whilst both got good scores and good user feedback, I have decided to bring the **Legacy Scale** forwards simply due to it's abiity to create connection and bring purpose to cooking.

ITERATIONS

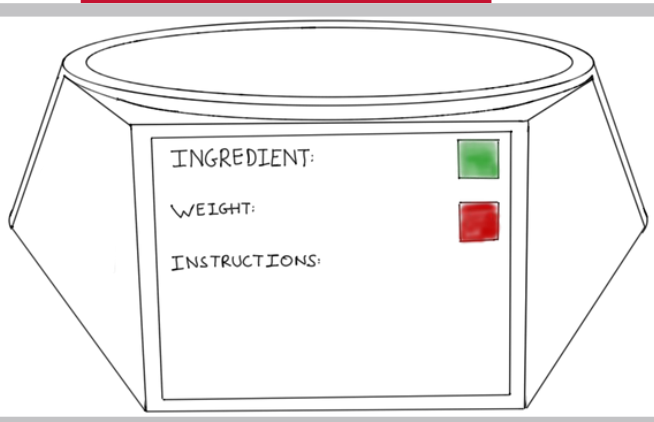
Francesco
23



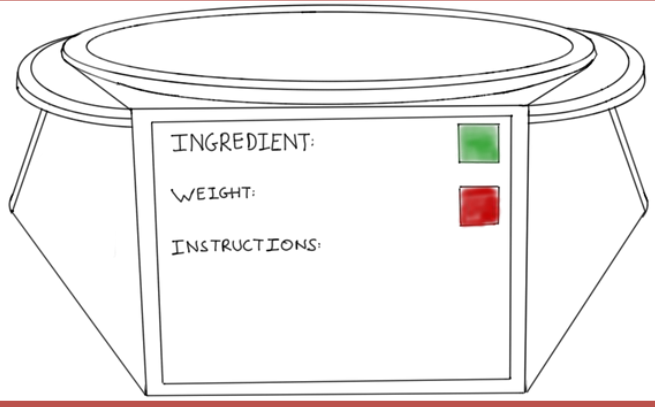
V1. The initial version has **analogue inputs** in the form of a knob and button and a simple **small screen** and LEDs as outputs to give the user the information they need.



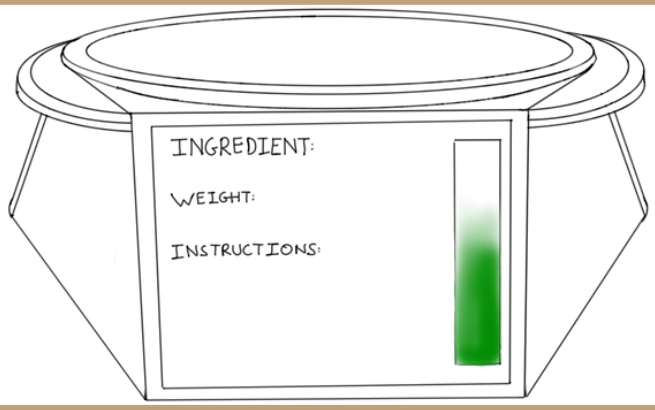
V2. This version replaces the inputs for a touch-screen which still gives the coloured outputs from the LEDs while also allowing for **bigger text** and space for **cooking instructions.**



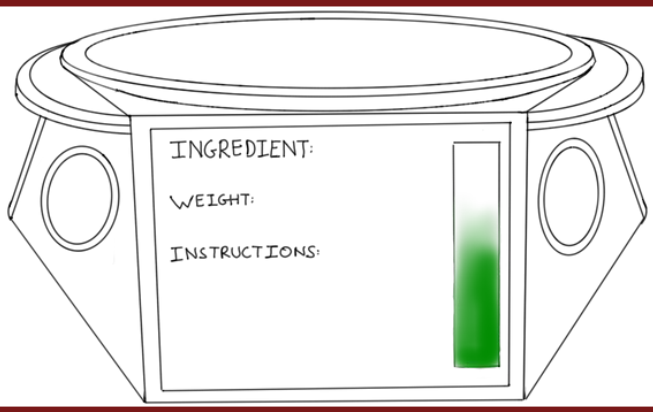
V3. The design was then smoothed in order to improve the **visual appeal** of the product.



V4. Handles were added to the sides of the scale to make it easier to lift up for a user and **manoeuvre** it. The wide area that they have ensures even those with low dexterity can lift it.



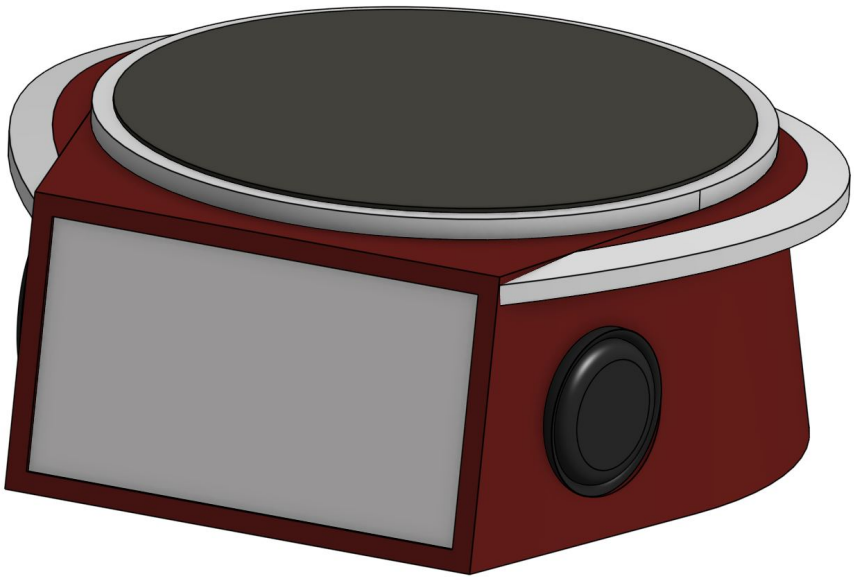
V5. Instead of the red and green light, a clearer scale was used through a **progress bar**. This shows the user how close they are to the wanted weight and can still alert them if they go over.



V5. To make the product easier to use, inspiration was taken by the **Voice-Controlled Microwave** and a speaker and microphone system was added to given further **inputs** and more **feedback**.

3D DEVELOPMENT

First 3D Design



The CAD on the left shows the final iteration in colour for the product. This was run by the users to gain feedback and work out issues they might have with it being:

"I struggle with touchscreens as is so I don't know if I could get this to work"

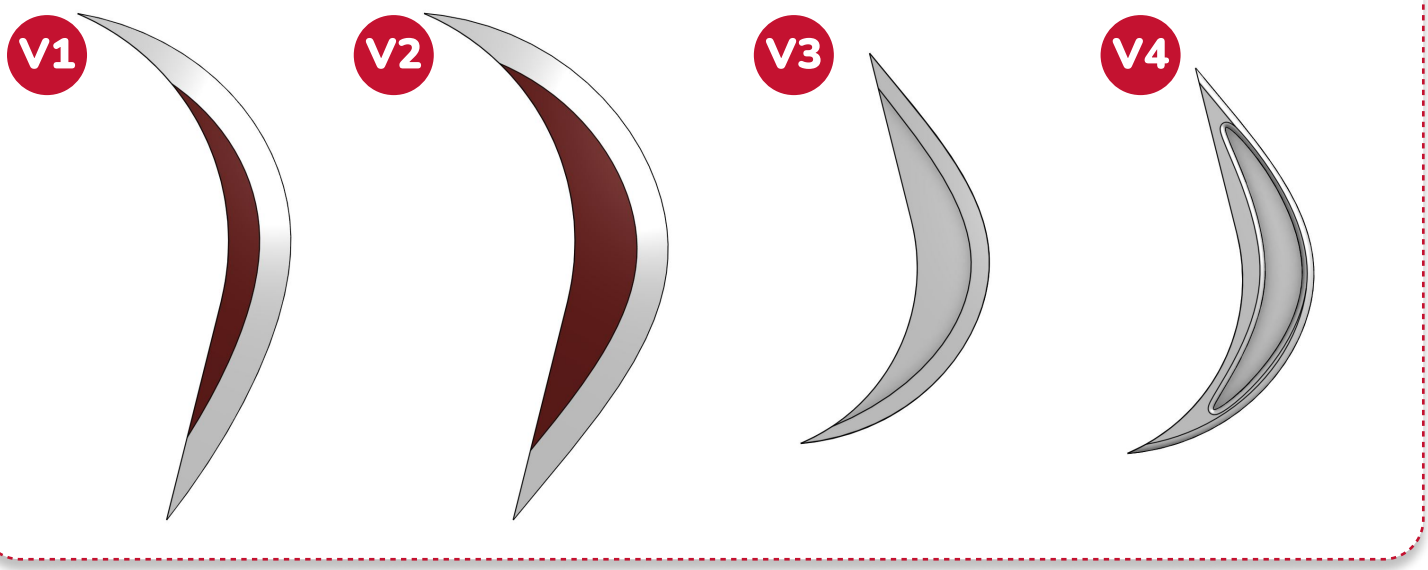
"I'm not sure if I want to speak to my scale"

"The handles look small there. Do you think I can hold them?"

To improve the design, different components needed to be split up and worked on in CAD separately to improve them for the final version.

3D Development

The Handles:



V1: This handle just follows along with the original design.

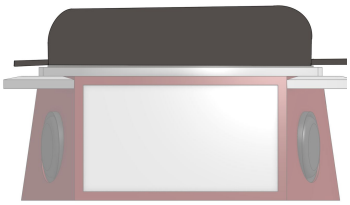
V2: Due to User Feedback, the handle's width was increased to give the user more area to grip.

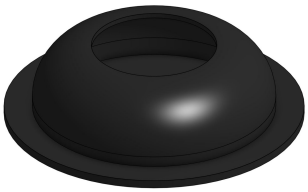
V3: A lip was added underneath the handle to make it easier to grip as it's a less smooth surface.

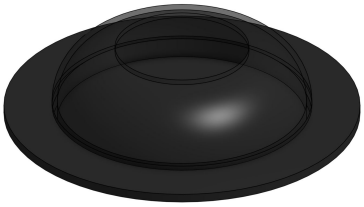
V4: The lip was smoothed out and rounded to make it more comfortable to hold, even for those with low grip strength and finger dexterity.

3D DEVELOPMENT

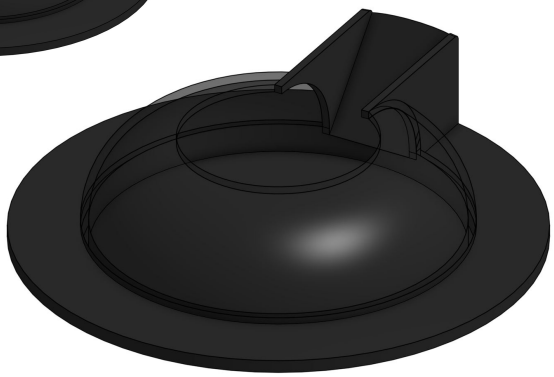
Adding A Bowl:



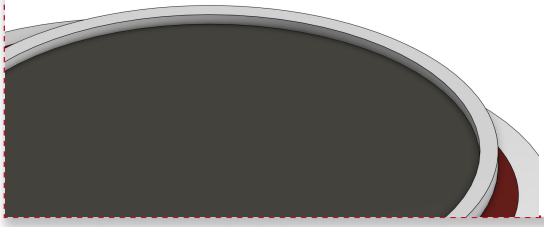




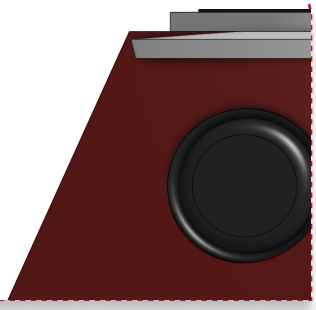
Having wanted to add a bowl to the design, thoughts were given to what it could look like, however, one source of feedback said the scale kind of looks like a face from the front on. Because of this, I decided to make it look like a top hat to add character to the design. Throughout the iterations, the brim was made larger to make it easier to hold, the top see through to see the ingredients, and finally a funnel was added to the back to pour out of.



Changes To The Base:



Two small, quality of life changes were made to the base. Firstly, an indentation, which can be seen on the left, was added to help hold onto the bowl without it slipping. On the other side, the angle of the screen was decreased in order to make it easier to read from a raised position.



Touchscreen Interface:

The User Interface needs to tell the user a few key pieces of information. Firstly, what the ingredient they need to add is. Then the weight that they are aiming for, which is complemented by the progress bar on the right which would turn green when that weight is reached, giving visual feedback. And finally instructions of what to do with the ingredients. In terms of input buttons, we have a forward and back button to go in-between steps and ingredients, a tare button on the right to zero the scale, and finally an audio button in the middle for auditory feedback.

FINAL PRODUCT

The Legacy Scale

Legacy is a smart scale that supports you and your family through the challenges that we all face in old age.

This product helps you carry on those traditional family recipes by intelligently recording them down. These can then be re-run through by the scale by both visually and vocally giving you instructions on ingredients, weights and steps to create all those family favourites

Francesco

LEGACY

Bringing family recipes forwards to the next generation.

THE PROBLEM

ELDERLY PEOPLE COOKING

Whilst searching for potential gaps in the current markets, we decided it was not necessary to look beyond the confines of our homes. It is known that elderly folk struggle with the handicaps imposed by old age. Arthritis and gradual muscular degradation impede or prevent the completion of physical goals whilst a wide variety of mental afflictions can add motivational issues to the mix.

From the ensuing interviews it was determined that Fatigue was the most prominent issue for the elderly. The three recurring problems were as follows:

Fatigue

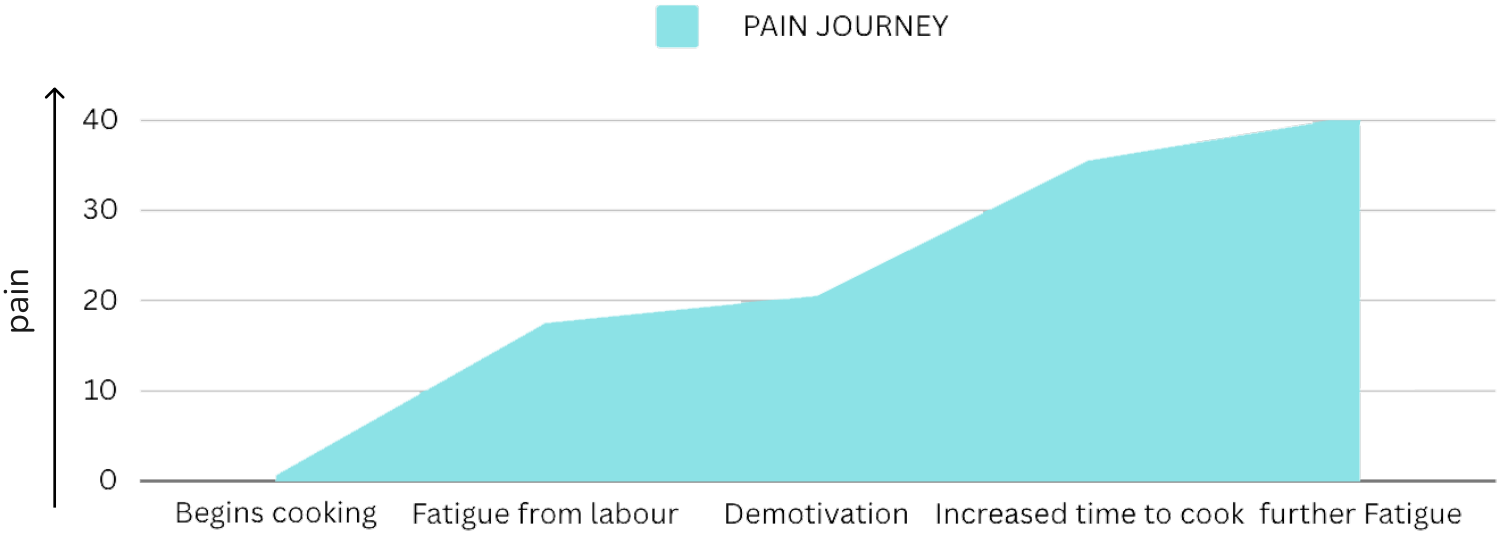
Fatigue is arguably the most pressing issue as it is almost completely ubiquitous amongst the elderly population. Weakening of the muscles is an entirely natural process and yet it is very rarely that specialised cooking equipment is used and the struggle is often written off as a common plight amongst older folk.

Forgetfulness

Mental afflictions - from minor amnesia to dementia - are less common than reductions in physical strength but are still prolific and problematic. These can make it difficult to follow recipes or even to difficulty with appetite and eating.

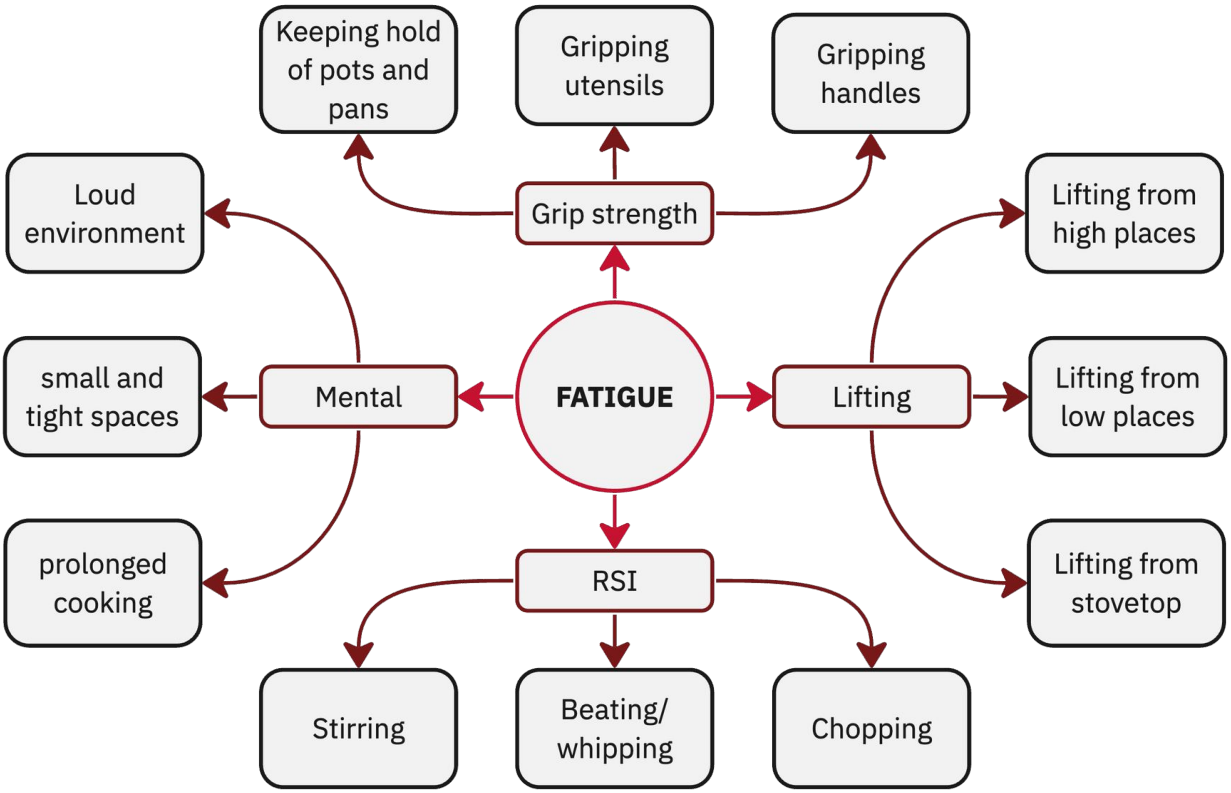
Loneliness

Loneliness is a far less explored affector of cooking and other everyday tasks faced by those who have lost loved ones or found themselves alone for other reasons



KEY INSIGHTS

Compilation of notable difficulties:



Insights and Opportunities:

Repetitive strain injury is a result of repeated cyclical movements. Such tasks are ideal for a machine to carry out due to their mechanical precision and durability.

Lifting, whilst being a difficulty, can be made significantly easier with improved handling design. Stability allows for more weight to be held since balance ceases to be an issue.

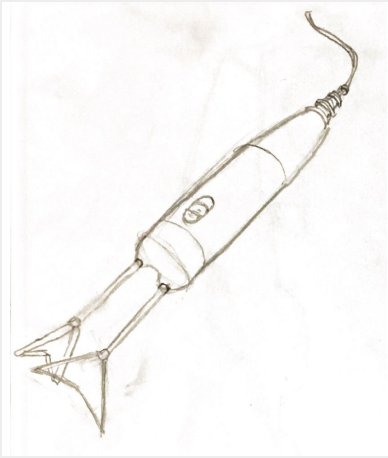
Grip strength ties in closely with lifting, but can also be assumed to always have time of hold as a factor. Prolonged strain can be made more comfortable with increased surface area to reduce force.

Mental fatigue ensues from attempting excessively complex tasks. Complex tasks can be made simpler with proper guidance or by optimising and reducing the task.

Mental stress will also follow if any task is done for too long. This can be alleviated by either introducing breaks or variety into the activities.

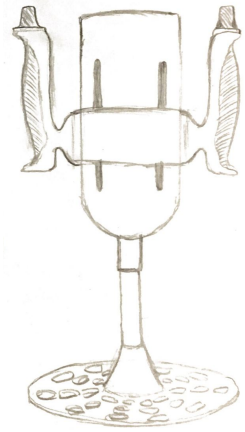
Lifting fatigue could also be prevented by optimising the workspace or tools so that lifting is reduced or even avoided all-together.

SKETCH IDEATION



EZ-WHISK

this device features two rotating whisks that are optimised to maximise ease of whipping or mixing anything from soups to mashed potato. Simple to operate and comfortable to hold. Long stems allow extended reach.

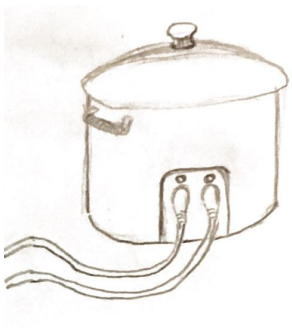


MASHOMATIC

The Mashomatic automatically jumps up and down, squashing any soft vegetables or fruit beneath it. The handles incorporate suspension so strain on the users wrists is minimal. Dual kill switches for safety.

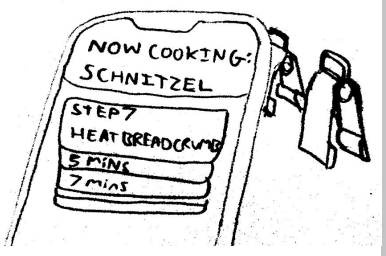
JAR POP-CAP

The Jar Pop-cap offers a simple alternative to regular jar lids with a snap fit lid that allows instant access to the jar without the chance of seizing closed. Fully recyclable and built to last.



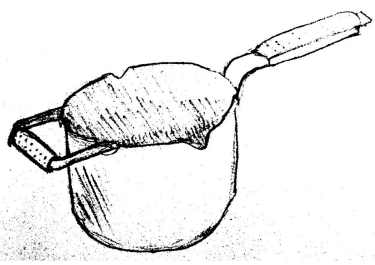
SYPHO-POT

A cooker that can be filled and drained without having to be hauled over to the sink. With the press of a button, soup can be pipe fed to a separate storage container and the pot can be filled with tap water for the next batch.



EVO-CHEF

Evo-chef is a system in which an app communicates with tags that are attached to the pots and pans to notify the person cooking when and what to do with light and haptic queues for a perfect, stress free result every time.



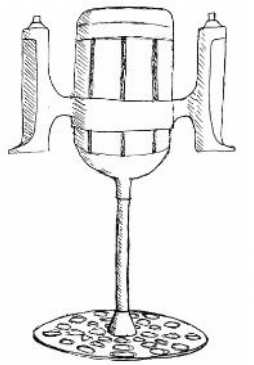
HANDIPOT

The HandiPot features a front handle to vastly reduce the difficulty of lifting. It also has two pouring grooves so that it is simple to pour contents out in a stable and controlled manner without getting burned.

PRIMARY SELECTION

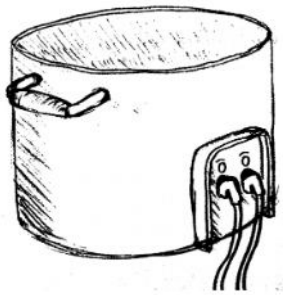
TOP 3 DESIGNS

MASHOMATIC



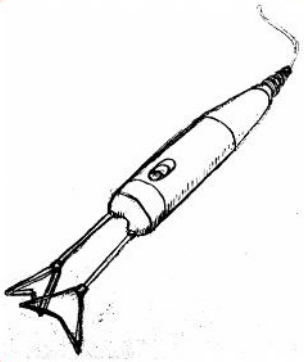
The mashomatic is both unlike anything on the else on the market today. It features a weighted mashing plate that is actuated to jump up and down cyclically while the user holds two floating handles to laterally guide the device and squash the food beneath it.

SIPHO-POT



The siphon-pot integrates a pump into a regular cooking pot to remove the need for any lifting whatsoever by transporting liquids to and from the stationary pot at the press of a button, both freezing and boiling substances.

EZ-WHISK



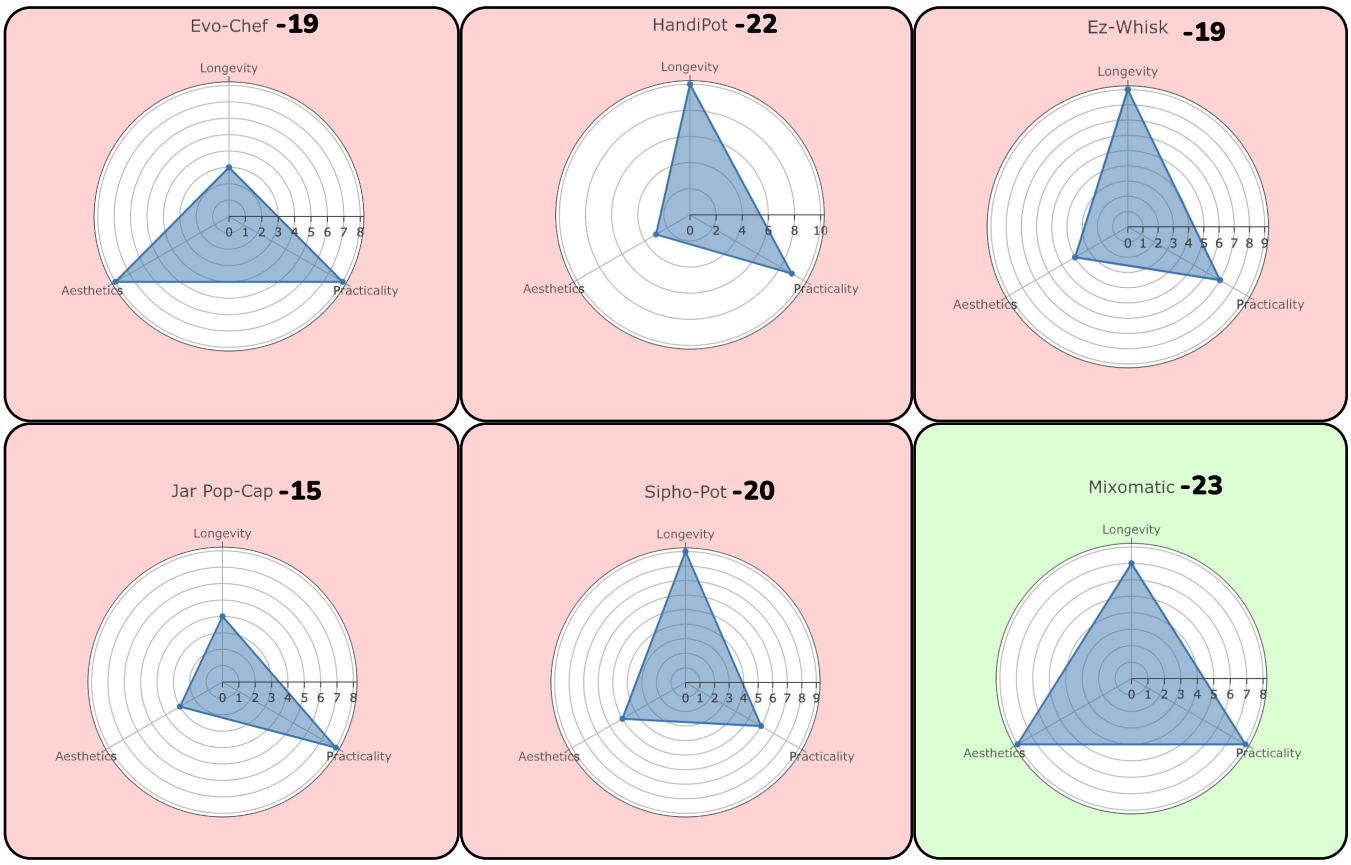
The EZ whisk fuses the technology of electric whisks and immersion blenders to bring to the user a highly effective and powerful whisk/beater which can reduce lumpy potatoes ands soups alike to a fine pulp. Blades are shaped for easy mashing in tight corners.

Of the six ideated concepts, the above three were deemed as the most feasible and viable within the team. The app integration and micro-electronics required for the smart tags in the Evo-Chef would prove highly difficult to implement. The HandiPot would be a proven and promising project except it lacks initiative and creativity. It simply does not stand out enough from available options to trailblaze in any respect. The same goes for the Jar Pop-Cap, which is simultaneously uninspired and possibly may itself get jammed onto an empty Jar hence negating its usefulness.

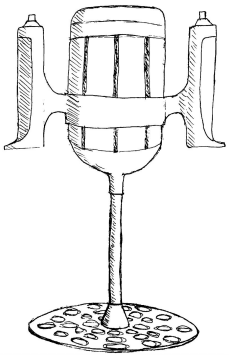
FINAL SELECTION

RADAR GRAPHS:

The below array of radar graphs maps out the appraisal of each of the concepts based upon user and team ratings of the superficial aspects of aesthetics, practicality and longevity. This was then combined with a more comprehensive gauging both of originality and feasibility. Each has a total score visible next to its name.



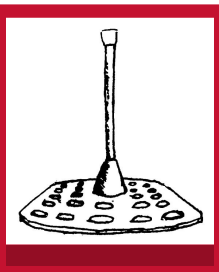
Mashomatic



After analysis and appraisal, the concept that rose above the rest was the Mashomatic. A very unique and intriguing design which features a weighted, jumping device that uses a flat, perforated masher to pulpify potatoes or other soft vegetables.

The masher concept holds potential in each of the three key categories of longevity, practicality and aesthetics. Being the most practically promising as well as entirely unique, the Mashomatic is the chosen concept for further development.

CONCEPT DEVELOPMENT

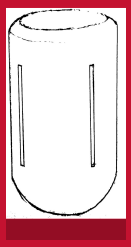


Angled Plate

A radiused square shape allows for better access in containers with corners when compared to a conventional circular plate.

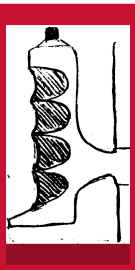
Mechanical Switch

A manual switch is more dependable than touchscreen, especially in moist and possibly dirty environments like domestic kitchens.



Reduced Slots

four slots rather than 8 for the handle module to slide on possibly means more friction, but also reduces the number of difficult-to-clean profiles in the main body cylinder.



Finger Grooves

Grooved rubber handles provide the user with superior grip and, by extension, better control of the device during operation and when moving it compared to smooth handles.

summary:

Multiple beneficial changes made to aspects of the design, both functionally and in terms of ease of maintenance.

3D DEVELOPMENT

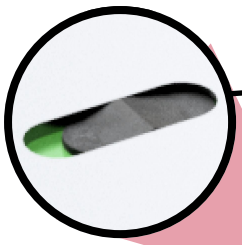
Final Design:

Material assignment:

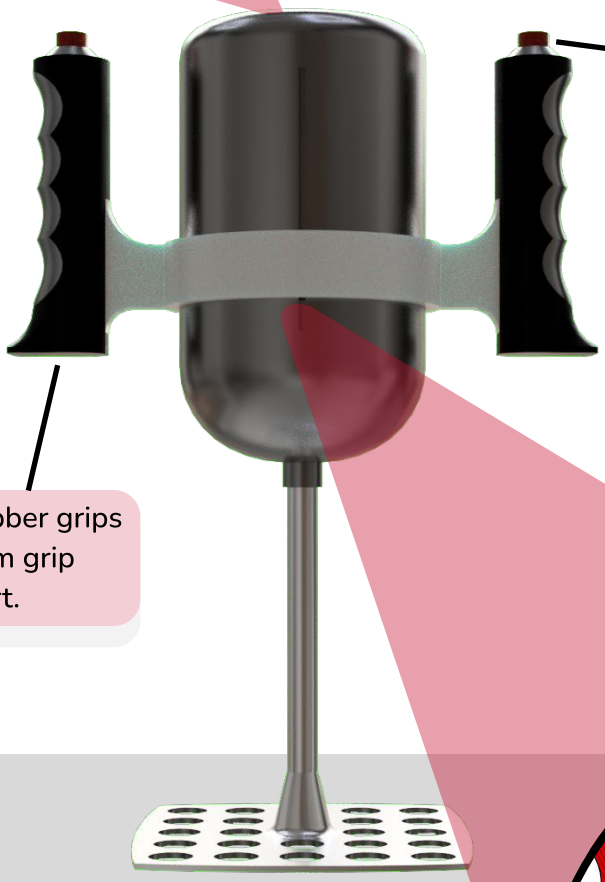
Aesthetically, a shiny finish on the stainless steel components has connotations with high quality of production whilst also looking polished and modern.
Vulcanised rubber on the handles is highly grippy and also soft making it comfortable to hold for extended periods of time. The top switch is also of rubber to prevent slips even when the user has damp fingers.

Practically, stainless steel is very hard wearing and has a reasonable amount of weight to it, so it was a sensible choice for the central weighted body. In order to prevent abrasion, the ring is made of ABS instead which should slide over the metal with little friction and not cause surface damage. The mashing plate is stainless steel for hygiene purposes as well as due to the high forces it is subjected to.

FUNCTION OVERVIEW



Manual power switch with indicative LED



Dual kill switches must both be compressed for operation to ensure proper and safe use.

Molded rubber grips for optimum grip and comfort.

A spring ‘floats’ the handles so that the users wrists are not jolted as the masher reciprocates.

Rack and pinion system used to power system. Gear and motor is attached to handles. Rack is attached to the cylindrical weighted body.

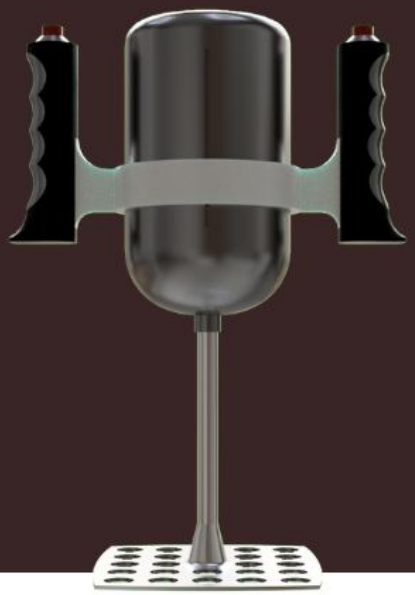
Gear raises winds the central column to the body to the uppermost position then releases it.

High torque motor powers a bevel gear that rotates the primary gear to ‘charge’ the next drop.

FINAL PRODUCT

MASHOMATIC

Sporting a powerful motor and spring suspension, the Mashomatic allows anybody to employ impressive force with great ease and effortlessly mash any soft fruits or vegetables in a matter of seconds. Ideal for older people but a welcome and unique addition to any modern kitchen.



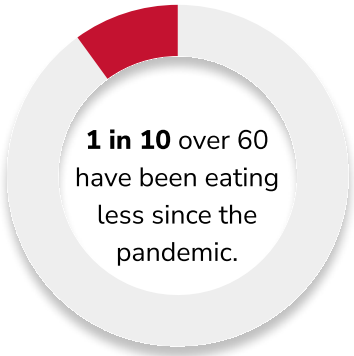
FINAL RENDER

INSIGHTS: User Reasrch

Sinai 38

Research Overview

This page shows a summary of the user research we conducted in order to analyse our user group. Our elderly population is increasing and those over 60 have face rising neglection since the pandemic resulting in them having to fend for themselves in their day to day lives but also in the kitchen.

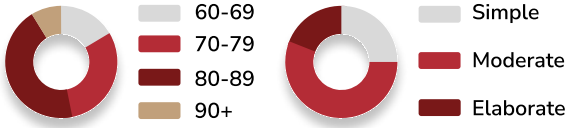


Nutrition is more important than ever particularly in older age. 1.4 million people over 60 have reported to eating less since the COVID-19 pandemic.

3.7M people

over 60 say they are now unable to eat healthy and nutritious foods in the UK

Survey Insights



'He eventually caved in with mixers and such when he got tired easily.'

'I think as she gets older she will not have the energy or the motor skills to be cooking as much as she is at the moment.'

'As grandparents age, they are less able to do more complex recipes, or forget their recipes coz usually they do them from memory.'

Interviews Insights

Birgitta Wiklund
A typical Swedish grandma, unfortunately been windowed. This led to her not enjoying cooking much any more.

'In the summers when daughter and grandchildren come it becomes more fun because you cook for more people.'

Patricia Pridmore
A retired academic who cooks for her less able husband. Having undergone knee surgery she has difficulty with general lifting.

As a result of extensive food preparation, 2 portions 3 times a day she often is left feeling frustration stemming from fatigue (particularly later in the day at dinner).

Key Insights

Lack of Enjoyment

Users can feel demoralised when cooking becomes a chore and necessity opposed to a fun activity, provided by company.

Forgetfulness

Users tend to face cognitive challenges effecting their memory for grocery shopping, and timeless recipes which can be lost forever.

Physical Strain

Likelihood of falls increases by 40% in people over 60 resulting in injury. This can make tasks in the kitchen very difficult.

How might we design cooking aids that reduce stresses for elderly users during common cooking tasks and also infuse purpose into their cooking process?

INSIGHTS: Market Research

Sinai 39

The two main markets which apply to our user group are Elderly and Assistive Devices and Small Kitchen Appliances. These were combine in the overview and product analysis.

Small Kitchen Devices

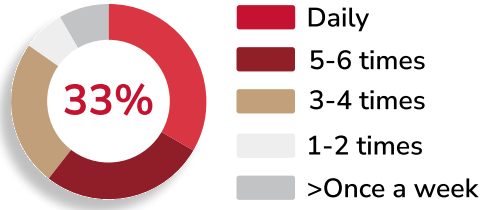
In 2024, the revenue in the **Small Kitchen Appliances** market worldwide amounts to

$133.6B

This is projected to grow from 2024 to 2028 with a **CAGR** of

%5.03

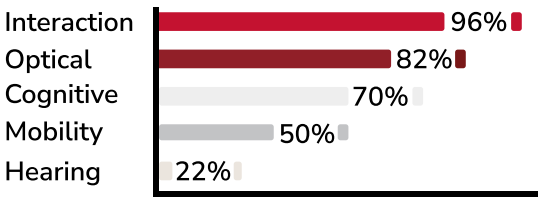
Frequency of people 60+ who cook



As people grow older they tend to become less able and as a result tend to stop cooking for themselves. With over 50% of elderly cooking almost daily it is imperative we continue to foster this independence.

Elderly and Disabled Assistive Devices

Percentage of people 60+ using different types of Assistive Devices



The majority of users suffer from a condition which requires an assistive device in their day to day life.

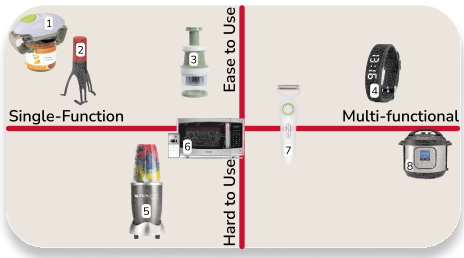
The global **Elderly and Disable Assisted Devices** market in 2022 was valued at

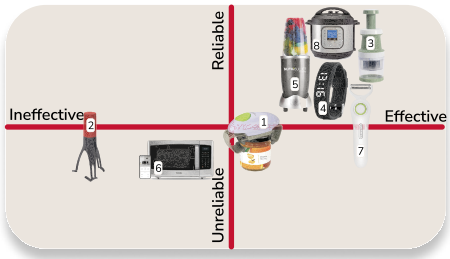
$27B

This is set to grow from 2023 to 2032 with a **CAGR** of

%5.70

Product Analysis





The product matrices put products against the following metrics, ease of use, functionality, reliability and effectiveness. The optimal product having all four positive attributes.

Key Insights

Multi-sensory Alerts

These can ensure users, regardless of sensory impairment, receive alerts. This helps us to cater to as many disabilities as possible.

Ergonomic Design

Tailoring products to accommodate different grip strengths ensures users with physical limitations, can use devices comfortably.

Discreet Design

Discretion without compromising functionality, allows users to feel comfortable using assistive products without feeling out of place.

Diverse Input Options

Large, tactile buttons reduce the complexity of devices as the elderly can struggle with small, touch-sensitive controls. Voice-operated input can sometimes be effective.

Multi-functionality

Offering modular attachments, a single device can serve multiple functions, expanding its utility and reducing the need for storage of more than one product.

Safety features

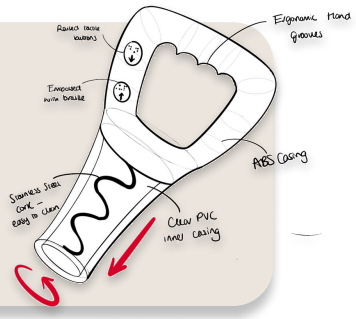
Integrating auto-off features and other kitchen product standards can greatly reduce the risk of accidents, helping to combat forgetfulness and cognitive impairments.

CONCEPT IDEATION: Initial

Sinai 40

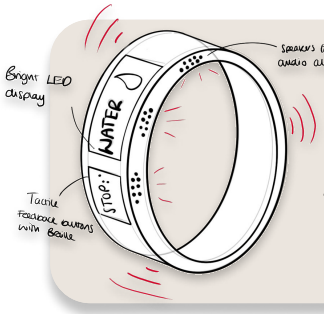
1 Wine Bottle Opener

The product is designed with an ergonomic handle for one-hand operation to, by a touch of a button, extracts the cork and by the touch of another, releases it.



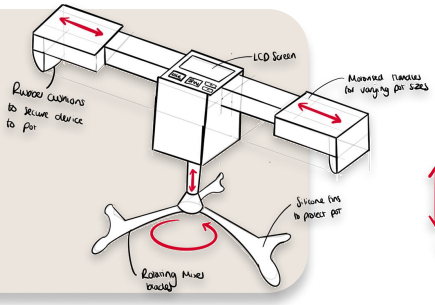
2 Alarm Reminder Watch

The watch can be programmed to remind the user to drink water and eat along with shopping lists and recipes. Alerts includes visual, audio and tactile.



3 Automatic Pot Stirrer

The product automatic clasps onto the edge of any pot or pan and automatically adjusts the height of the blades to stir the contents at your desired speed.

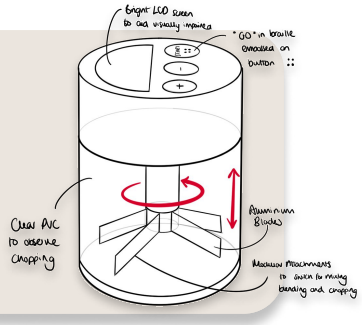


4 Hydration Reminder

Set alerts in regular interval for you elderly family member to stay hydrated. Additionally send either automated or pre-recorded words of motivation.

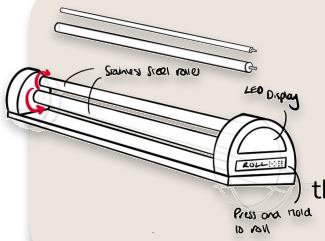
5 Electric Blender

The product allows for both rotation of the blade and vertical movement to either chop or mix ingredients, or even both, with a range of attachments.



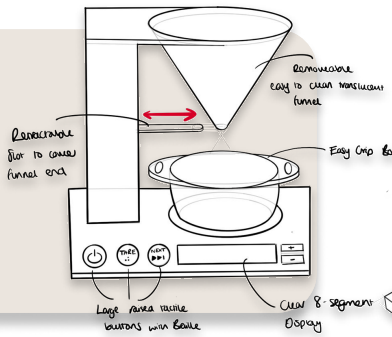
6 Electric Pasta Maker

This product can be used for both pastry and pasta, rolling and cutting dough to varying thicknesses to depending on the attachment used.



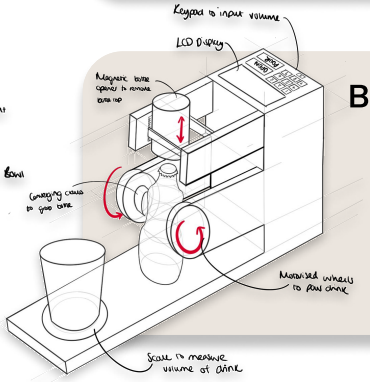
7 Automatic Scale

Either upload a recipe to the device or manually input your desired measurement of your ingredient and the slot will retract to release the amount.



8 Bottle Opener and Pourer

By the push of a button have you ice cold beverage opened. Input the amount you want on the keypad and it stop pouring once the amount is reached.



User Feedback

Anthony Walsh

A traditional Jamaican grandfather making his way back into the kitchen after having returned home from battling cancer.

- 4 I do forget to drink water from time to time. ✓
- 5 I do sometimes line a drink so this would be nice. ✓
- 7 I don't really use measurements when cooking.
- 8 When I chop onions my eyes water so I like this. ✓

Pearletta Livingstone

A retired Jamaican grandmother who enjoys cooking and baking for her husband, children and nieces and nephews at family gatherings.

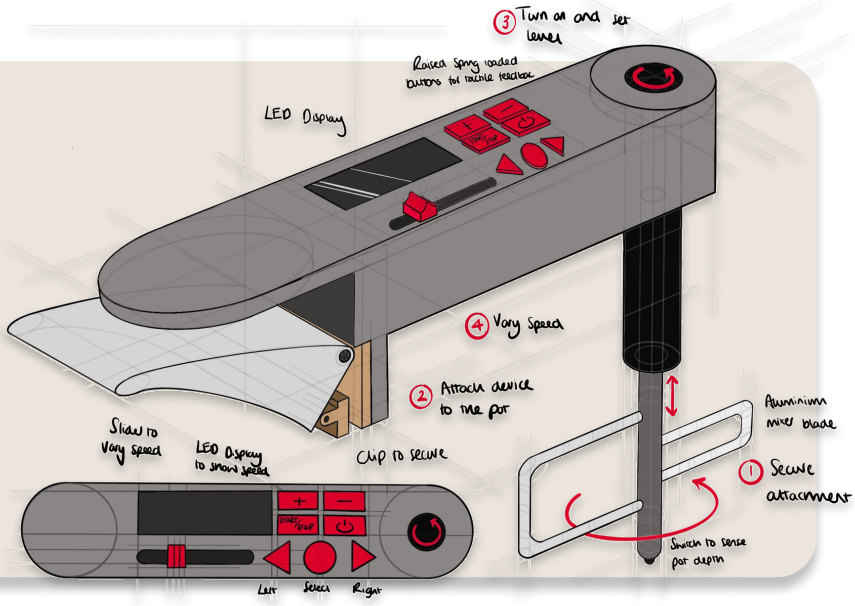
- 3 Looks like a nice idea but a bit cumbersome. ✗
- 4 Would be great for someone like my mum (84). ✓
- 5 Would really speed up prep for a big meal. ✓
- 6 I could really do with this for pastries. ✓

CONCEPT IDEATION: Developed

3 Automatic Stirrer

A kitchen aid designed to offer hands-free stirring for a variety of dishes, ensuring consistent stirring without physical strain.

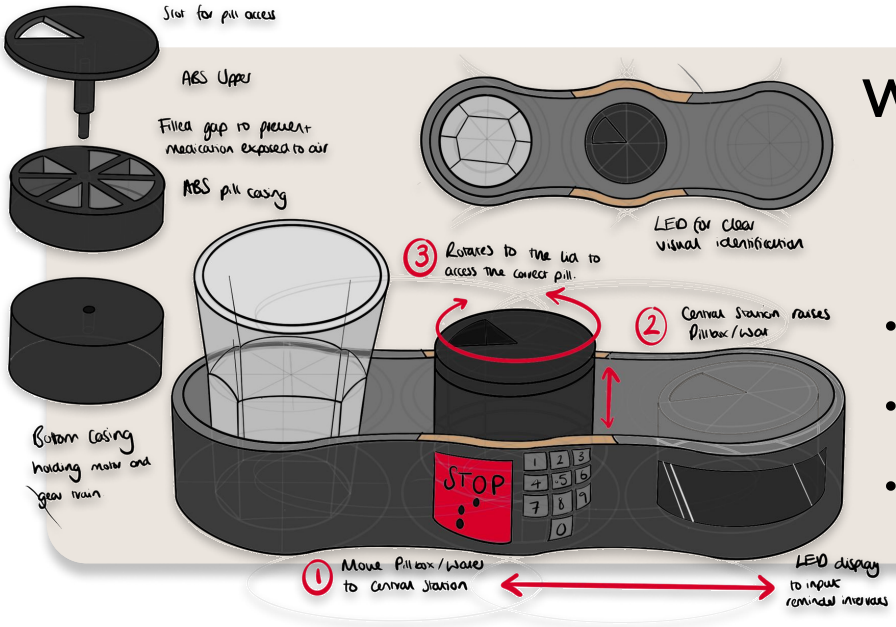
- **Accessibility**: Spring loaded buttons with tactile feedback with a slider to vary the speed.
- **Personalisation**: Pre-save set depths of different pots and speeds.
- **Modular Attachments**: Vary attachments to accommodate for different viscosities and different stirring patterns.



4 Water and Pill Reminder

The Water and Pill Reminder device is designed to aid the elderly with hydration and medication through timely alerts by moving and raising the cup at through the central cylinder.

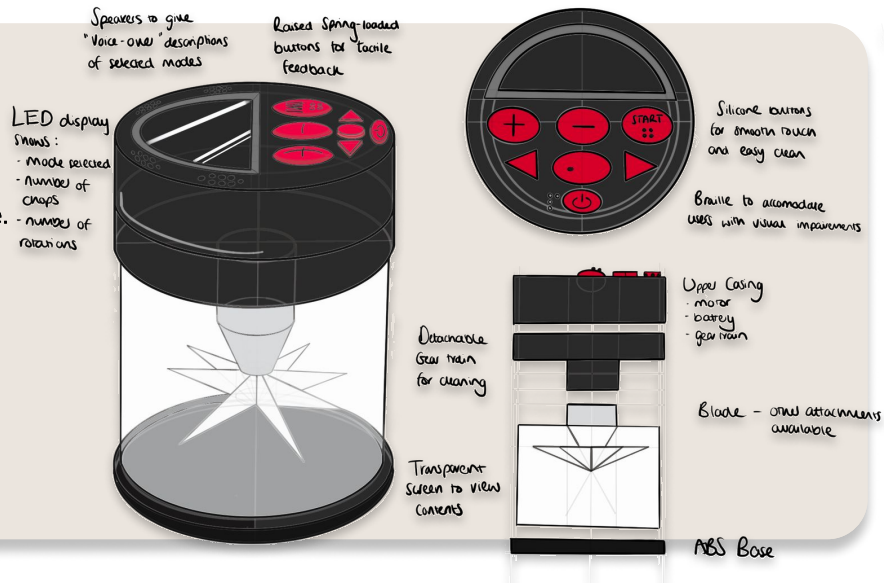
- **Accessibility**: Keypad and buttons complete with Braille for visually impaired.
- **Personalisation**: Input your desired reminder interval and upload pre-recorder voice memos.
- **Smart Connectivity**: Potential for app operation by a carer to set alarms for their dependent



5 Electric Mixer

This device simplifies the process of blending with accessible controls and smart features that ensure a safe and convenient kitchen experience.

- **Accessibility**: Clear, large-button interface to with audible and visual alerts depending on mode to accommodate a range of impairments.
- **Personalisation**: Programmable settings to remember favourite blends and chops.
- **Safety**: Automatic shut-off for and anti-slip base to secure the device during operation.



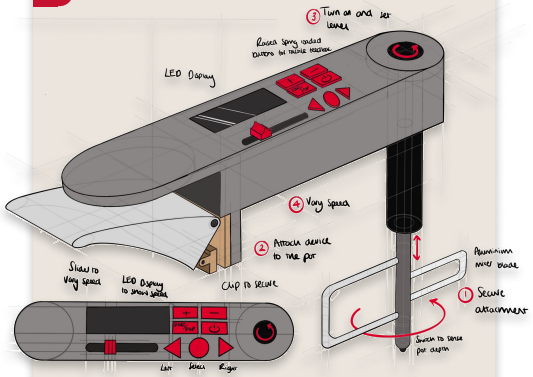
CONCEPT IDEATION: Validation

Sinai 42

Clay Muhammed

Clay is NHS Carer assigned to Anthony after having come out of hospital. He visits Anthony 3 times a day to prepare each meal and visits over 10 people every day and has been a carer for almost 10 years. I have used his feedback to aid my decision in pursuit of my final product.

3 Automatic Stirrer



Feasability
Desirability
Innovation

Positives

"Would be useful for residents who enjoy cooking but need extra help."

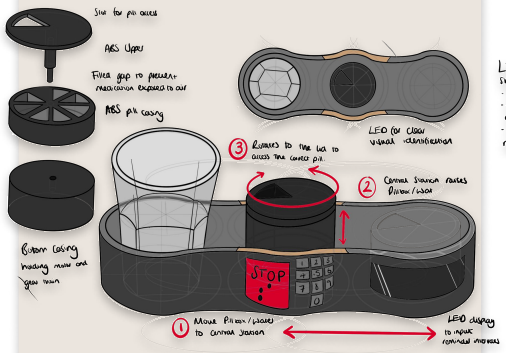
"Different speed settings catering to various dishes... quite practical."

Negatives

"I'll have to make sure it's secure so it doesn't fall in the food "

"Needs to be robust enough to stir hearty meals without coming loose."

4 Water/Pill Reminder



Feasability
Desirability
Innovation

Positives

"...Would be incredibly helpful for maintaining routines."

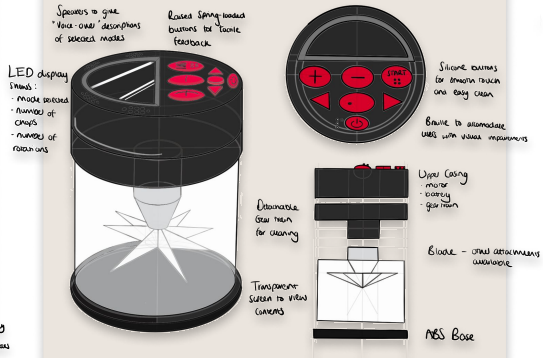
"Could take some pressure off the staff in terms of monitoring."

Negatives

"Essential that it has a long-lasting battery for continuous use."

"I really like it but I'm not sure it helps them in the kitchen."

5 Electric Mixer



Feasability
Desirability
Innovation

Positives

"If the display is as clear as you say, it could really help people with poor vision."

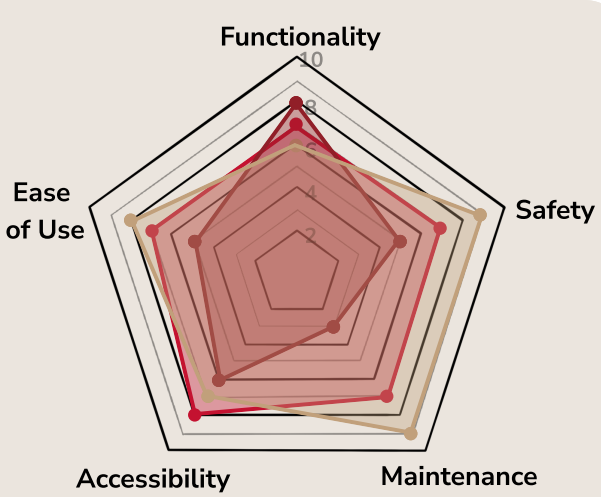
"So versatile...could make everything from smoothies to soups."

Negatives

"Multiple buttons might complicate things for some.

"Is it suitable for those with limited hand strength?"

Evaluation



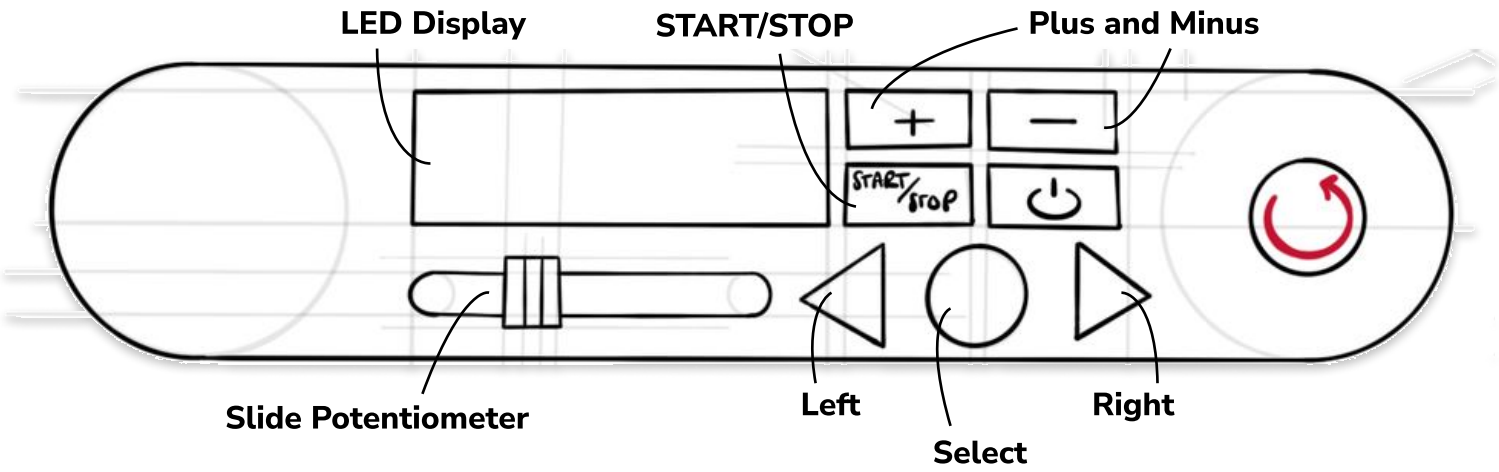
"... the water/pill reminder is my favourite... it helps the person and me out. I'm scared about the blender, cleaning might be a bit dangerous, but the automatic stirrer would really help people in the kitchen. "

After Clay's insight the product I am pursuing is the Automatic Stirrer. Despite the water/pill reminder being his favourite, it doesn't target our user group specifically in the kitchen. Insights to consider include:

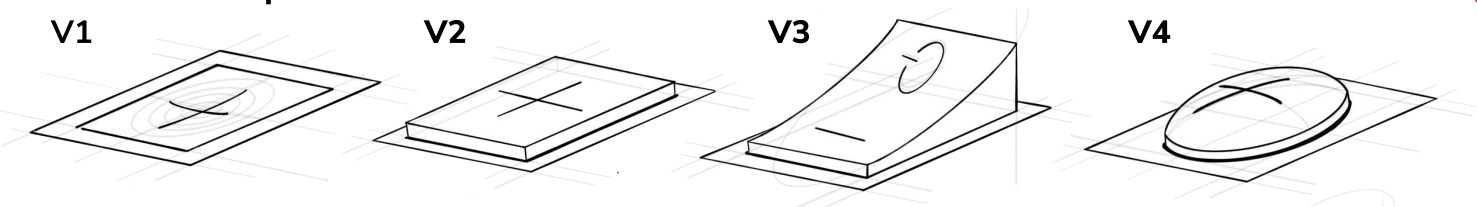
- **UI**: Buttons should not only large and clear but kept to a minimum.
- **Grip**: Ensure it is suitable for those with limited grip strength.
- **Security**: The stirrer must be robust enough to cope with more viscous liquids and not get lose while still being easy to detach.

CONCEPT DEVELOPMENT

User Interface



Button Shape



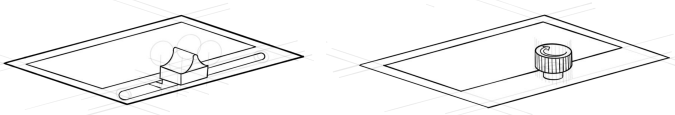
The Subtle Press
A sunken in button with a minor spring-back. The groove is sunken in to match the impression of a finger.

The Heavy Offset
A large raised button with a strong rectangular profile. Silicone material as seen commonly on TV remotes.

The Rocker
A rocker switch provides a clear definitive sign whether the device is on/off with a satisfying user experience.

The Circle Click
An oval shaped clear acrylic button with a smooth finish providing a gentle but responsive click

Potentiometer



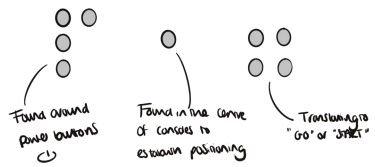
Slider
A sleek design, providing a definitive visual cue of the speed. Its curved edges and smooth glide action allows for precise adjustments.

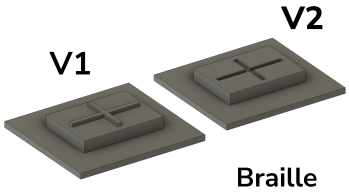
Rotary Switch
The dial offers an intuitive and satisfying twist action for users to adjust speed. Its size reduced space and ridged edge design ensures a secure grip.

Button Tactility

Embossing

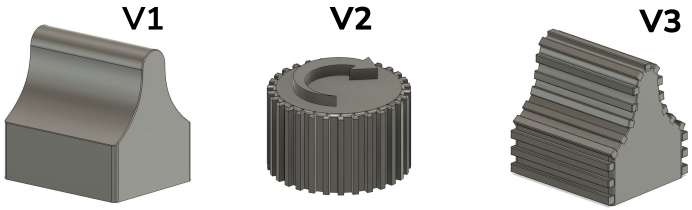
Buttons have been embossed (V2) opposed to debossed (V1) as it chosen when writing braille for faster symbol identification.





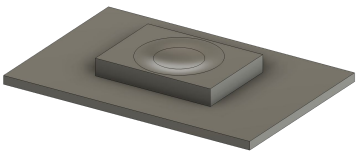
Braille

Braille and shaped buttons allows the product to be accessible to the blind and visually impaired.



With users commenting and giving positive feedback on both designs, they were combined, taking the ridged edge of the rotary switch and the slider mechanism.

Having gained some feedback from users the most optimal button combination was V1 + V2. Some of the feedback and insights about the designs and button choices include:

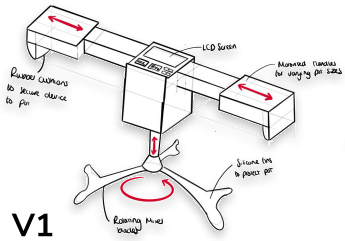


CONCEPT DEVELOPMENT

Grip

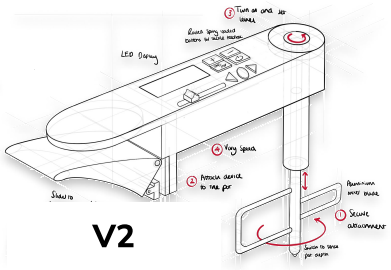
The grip is highlighted as a major feature of the product. It needs to me secure to the main body and also be accessible to those with varying grip strengths. Over 50% of people over 60 suffer from arthritis meaning it is imperative that the handle is comfortable and

Shape Design



V1

The original design was a concept where motors were retracted the arms to the sides of pot but was *"...too cumbersome"*.



V2

The updated design occupies half the radius of the pot opposed to the full diameter reducing size, resulting in an easier storage.

Handle Design

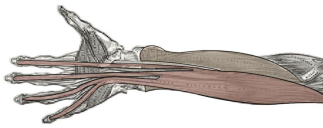
Horizontal

Vertical

Assuming the user is standing at the kitchen counter the handle, the vertical handle activates the lower forearm. Whereas the horizontal allows full activation of the providing greater overall strength

Vertical

Horizontal



Handle Shape

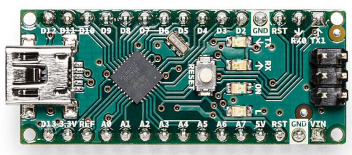
A circular handle appeared sleek effective design at first, as seen above, but the ergonomic solution as applied in many day-to-day products appears far superior. A handle moulded to the shape of the users hand will provide added comfort and alleviate any additional pressure needed to operate the clamp.

Technical Development

Infrared Temperature Sensor

The infrared sensor is a superior alternative to a physical thermocouple as it will not come into contact with the food but accurately report the temperature.

Arduino Nano



The compact nature of the Arduino Nano complete with digital and analogue I/O makes a perfect choice for varying the speed and quickly process temperature changes.

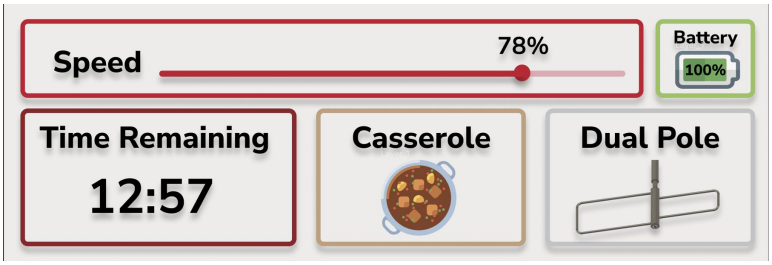
Button Mechanism

Latching Button

This mechanism allows the button to be inset when pressed and once pressed again raised back to its original position. This provides a clear indicator to show whether the button has been pressed.

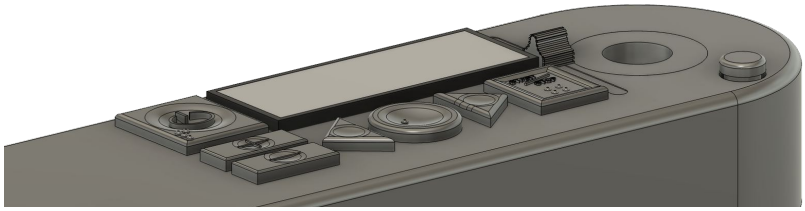
FINAL CONCEPT: Development

LED Screen



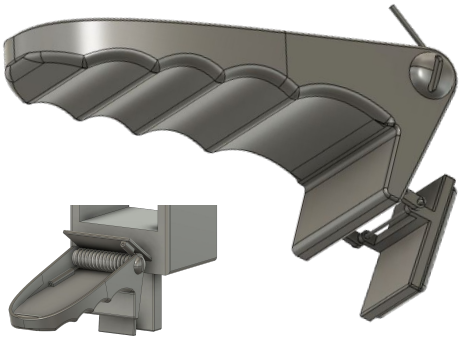
LED display is detailed through the image above. The display clearly illustrates the speed the attachment is rotating at, the battery percentage, stirring time remaining, and both the type of blade and meal being prepared.

User Interface



The user interface saw a rearrangement as a result of the feedback. The design of V1 and V2 were combined as shown with filleted edges to produces a smooth sleek finished. A combination of latching and simple spring loaded b

Ergonomic Grip



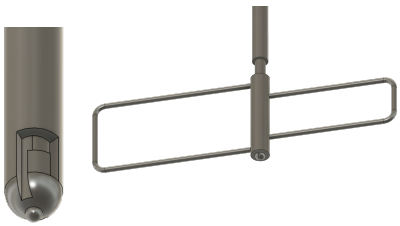
The handle aims to mimic the hand imprint on the putty. These grooves help to improve both comfort and grip to limit unnecessary strain on the users hand. This is also facilitate by a spring clamp system with a low spring coefficient to allow an easy yet secure clamp.

These bumps seen on the upper piece of the thumb groove further aid grip and prevent slipping when applying attaching the device to the side of a pot.

Clamp

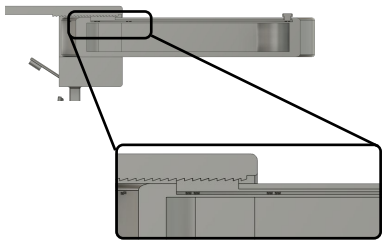
The clamp operates on a system with two pivot points, suitable for all pot shapes and sizes. Either surface is complete with a silicone pad; the ideal material being heat resistant and prevent and flexible not to damage kitchenware.

Attachment



The inner sleeve of each attachment features notches which slot into the mini actuator extending from the the main body. The 70° clockwise rotation helps to lock the blade in place to cop with stirring thicker liquids.

Pot Size Adjustment



To vary the distance of the blade from the pot depending on the pot size a push button spring is attached to a pair of compliment-ing teeth which retract and lock into place, dictating the size of the radius which the blade will rotate.

FINAL CONCEPT: Exploded

Electronic Casing

2.8" TFT LCD Screen
Display to clearly illuminate the user interface

Latching Buttons
Providing tactile feedback to carry out functions.

Push Latch
To adjust the diameter of the stirrer.

Upper Casing
The upper half of the electronic housing.

DC Motor
To operate the main functionality, stirring a pot.

2000 mAh Rechargeable Battery
To power and recharge the device.

Arduino Nano
To easily control the motor speeds and take frequent temperature readings.

Lower Casing
The lower half of the electronic housing.

Miniature Actuator
Varying length to control the depth of the attachment depending on that of the pot.

Stirring Attachment
To mix and stir the contents of the pot.

Push Button Switch
To trigger the miniature actuator to retract, preventing the blade from scraping the bottom of the pot.

Clamp

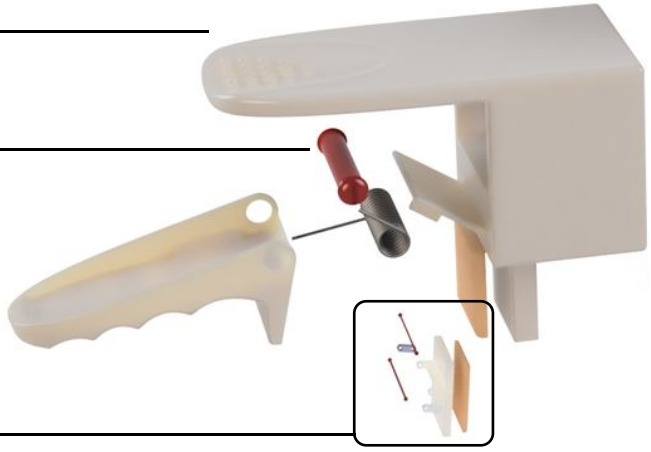
Upper Handle
Where the thumb is placed to clamp and secure the device to the pot.

Screw
To secure the spring in place.

Spring
To facilitate the clamping mechanism.

Ergonomic Handle
To allow provide comfortable grasp on the users fingers when clamping.

Variable Clamp Teeth
Dual pivot joint mechanism to allow freedom to clamp pots at varying angles.



FINAL RENDER

STIRMASTER PRO

Summary

A revolution in kitchen convenience, the StirMaster Pro offers an effortless solution to continuous stirring, ensuring even heat distribution and texture in your culinary creations. Designed for hands-free operation, this device easily attaches to your pot, with an adjustable arm to fit a wide range of sizes. The durable, food-safe materials promise a long-lasting addition to any elderly chef's arsenal, while the sleek design adds a touch of modernity to the kitchen. With variable speed settings controlled via an intuitive interface, the StirMaster Pro caters to a variety of dishes, from delicate sauces to robust stews, ensuring perfect results every time .